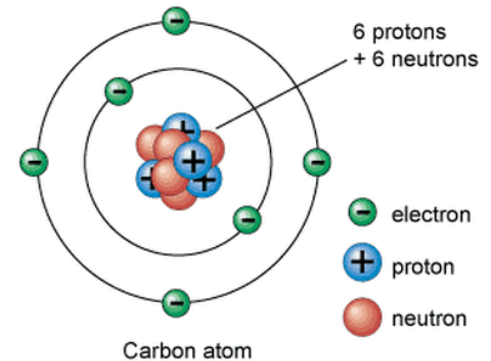
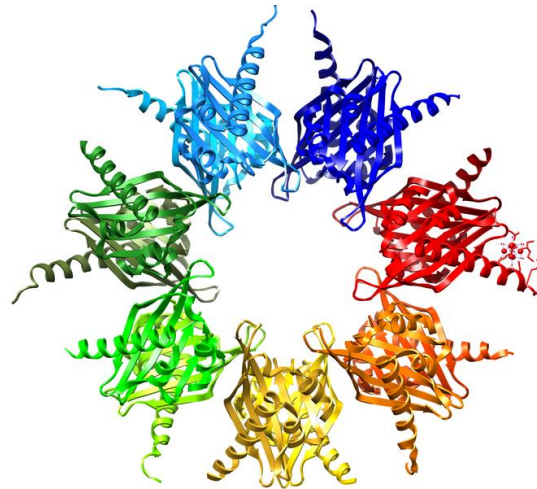
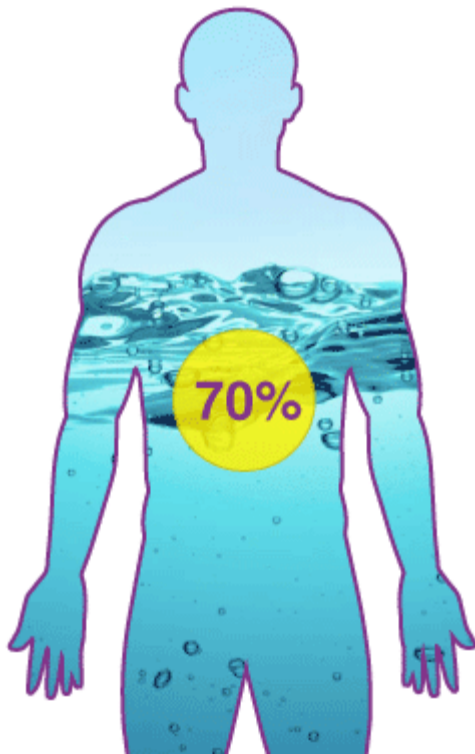


Chemistry of Life

(Chemical foundation)

Lecture 3-5



Life depends on chemistry!

- When you eat food or inhale oxygen, your body uses these materials in chemical reactions that keep you alive
- Just as buildings are made from bricks, steel, glass, and wood, living things are made from chemical compounds
- Wouldn't you want an architect to understand building materials? Same idea applies to geneticists, ecologists, zoologists, botanists, biologists, and etc.

let's learn from a real-life
example
why the chemistry of life
is important...

The main elements that compose the human body molecules (including water)

Element	Symbol	% Mass	% Atoms
Oxygen	O	65.0	24.0
Carbon	C	18.5	12.0
Hydrogen	H	10	62.0
Nitrogen	N	3.2	1.1
Calcium	Ca	1.5	0.22
Phosphorus	P	1.0	0.22
Potassium	K	0.4	0.03
Sulfur	S	0.3	0.038
Sodium	Na	0.2	0.037
Chlorine	Cl	0.2	0.024
Magnesium	Mg	0.1	0.015
All others		< 0.1	< 0.3

Atomic model

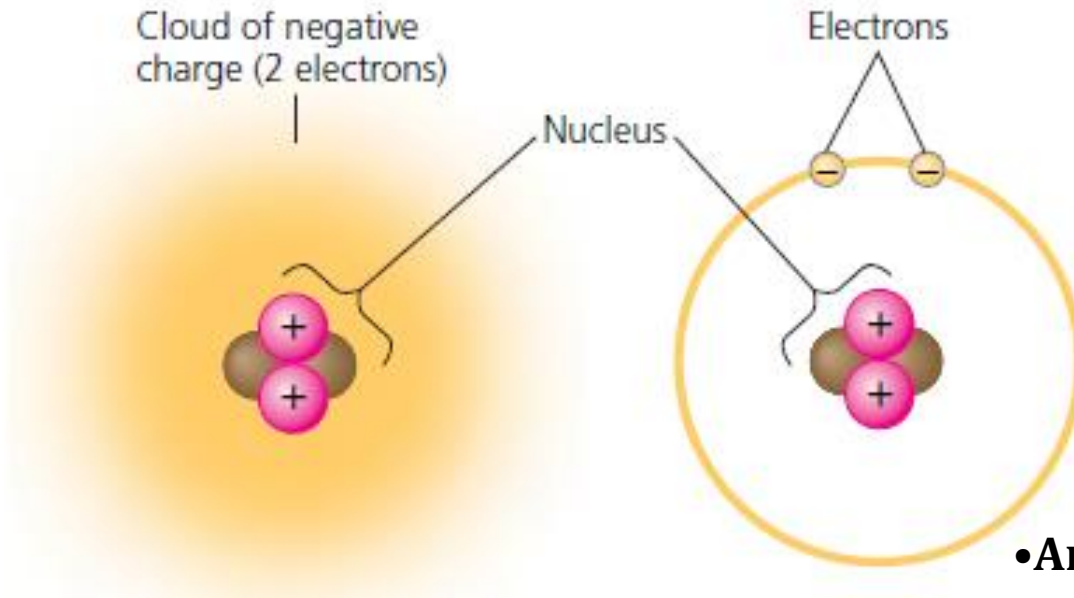
The concept about the structure of atom by different scientist. Most important are:

1. Thomson's Atomic Model
2. Rutherford's Atomic Model
3. Bohr's Atomic Model
4. Modern Atomic Model – Quantum mechanical model

Atomic Structure

Atoms are neutral

Numbers of e^- and p^+ are the same in one atom



•A nucleus

- *small, heavy part of the atom*

Nucleus contains:

- **Proton** - +ve charge, All atoms are distinguished by the number of protons it has (**atomic number**)
- Neutrons – NO charge, Have same mass of protons

•An **electron cloud**

- *large, lightweight part of the atom*
- *- ve charge, determine bonding properties of a substance.*
- *orbit the nucleus of the atoms*

Number of proton is fixed for an atom of an element

Proton number and nucleon number

The proton number (Z)/ atomic number → number of protons in the nucleus of an atom *of an element*. The Periodic Table of elements is arranged in order of the proton numbers of the individual elements. The **chemical identity of an atom** can be **determined solely by its atomic number**.

- The **nucleon number (A)/ mass number** → number of protons + neutrons in the nucleus of an atom of an element.
- number of neutrons = nucleon number – number of protons
$$= A - Z$$

For example, an atom of aluminium has a nucleon number of 27 and a proton number of 13. So an aluminium atom has $27 - 13 = 14$ neutrons.

THE ATOM: BASIC STRUCTURE

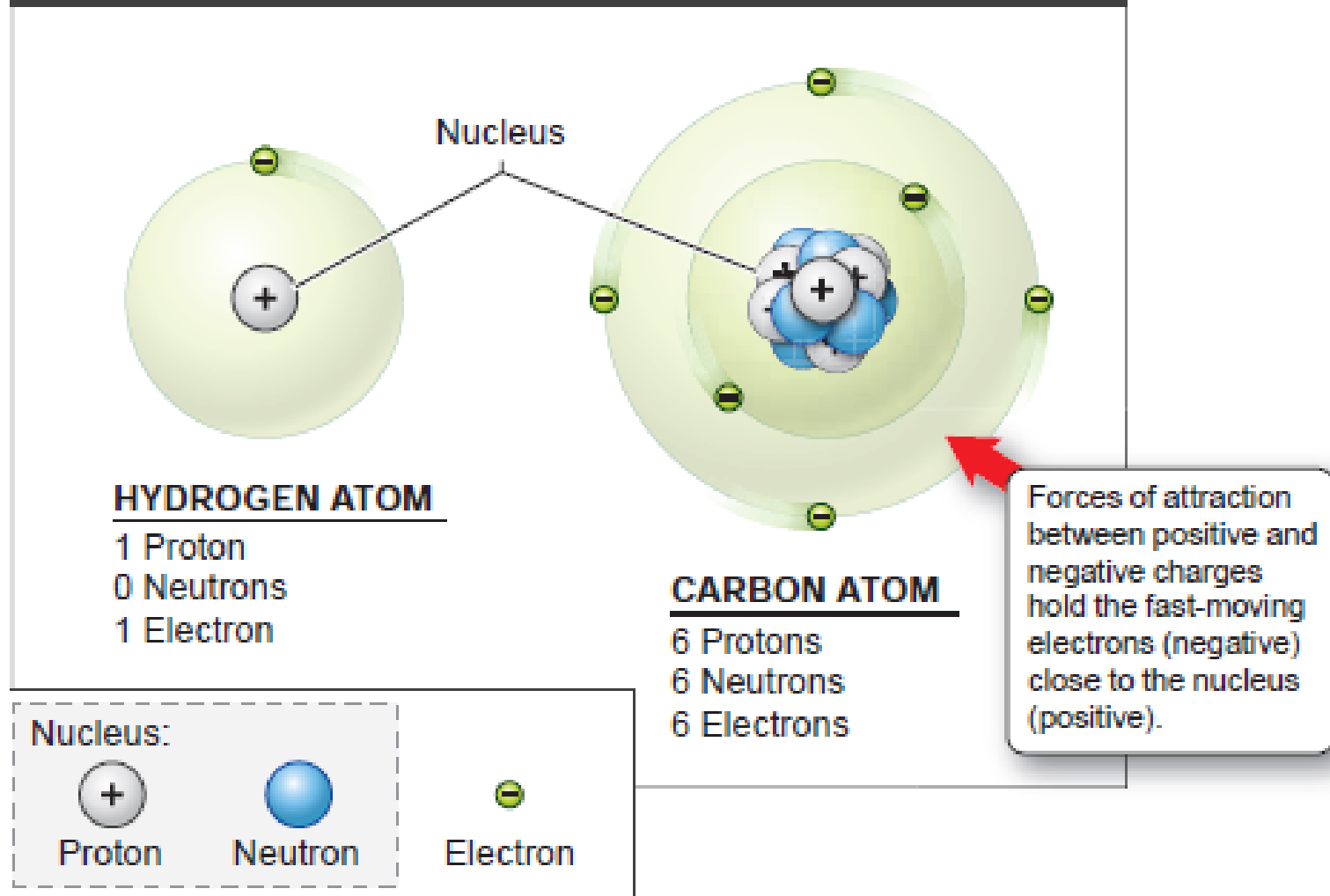
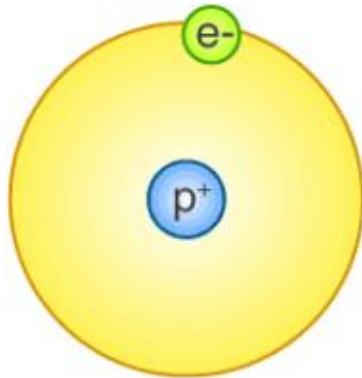


FIGURE 2-2 The atom. At the center of an atom is a nucleus containing protons and (in all elements except hydrogen) neutrons. The nucleus is surrounded by electrons whirling about in a cloud.

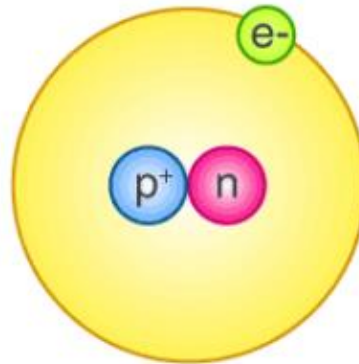
Isotope

Same number of protons but
different number of neutrons

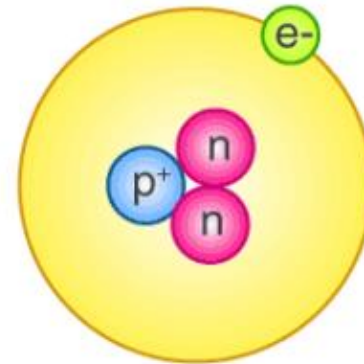
Hydrogen Isotopes



Hydrogen-1
Mass number : 1



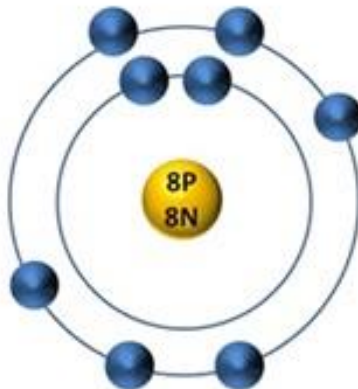
Hydrogen-2
Deuterium
Mass number : 2



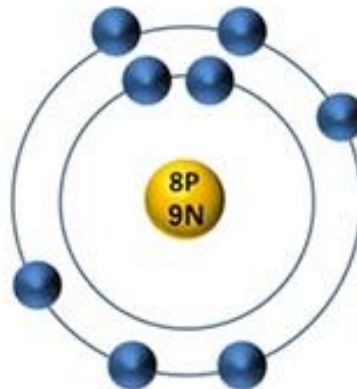
Hydrogen-3
Tritium
Mass number : 3

Examples:

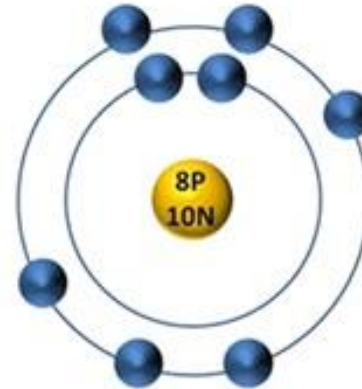
Oxygen Isotopes



¹⁶O Isotope



¹⁷O Isotope



¹⁸O Isotope

ISOTOPES-Differ in number of neutrons

Atoms with the same atomic number (i.e., the same number of protons) but different atomic mass are isotopes.



CARBON-12

CARBON-13

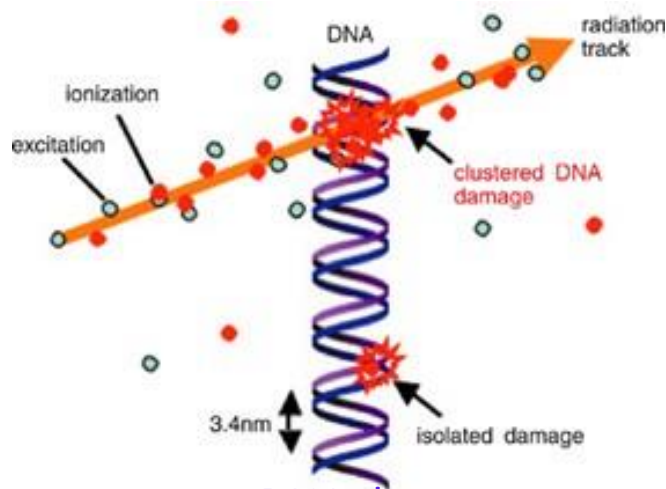
CARBON-14

⊕ Protons	6	6	6
● Neutrons	6	7	8
• Electrons	6	6	6
Atomic mass	12	13	14

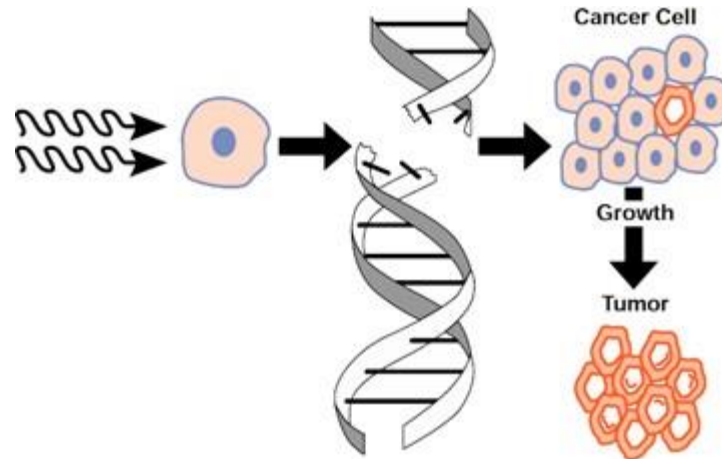
FIGURE 2-4 Isotopes are atoms that have extra neutrons or missing neutrons.

Isotopes are different forms of the same element. Isotopes are atoms of the same element with different mass numbers.

Isotope in biology



DNA damage
Free radical formation



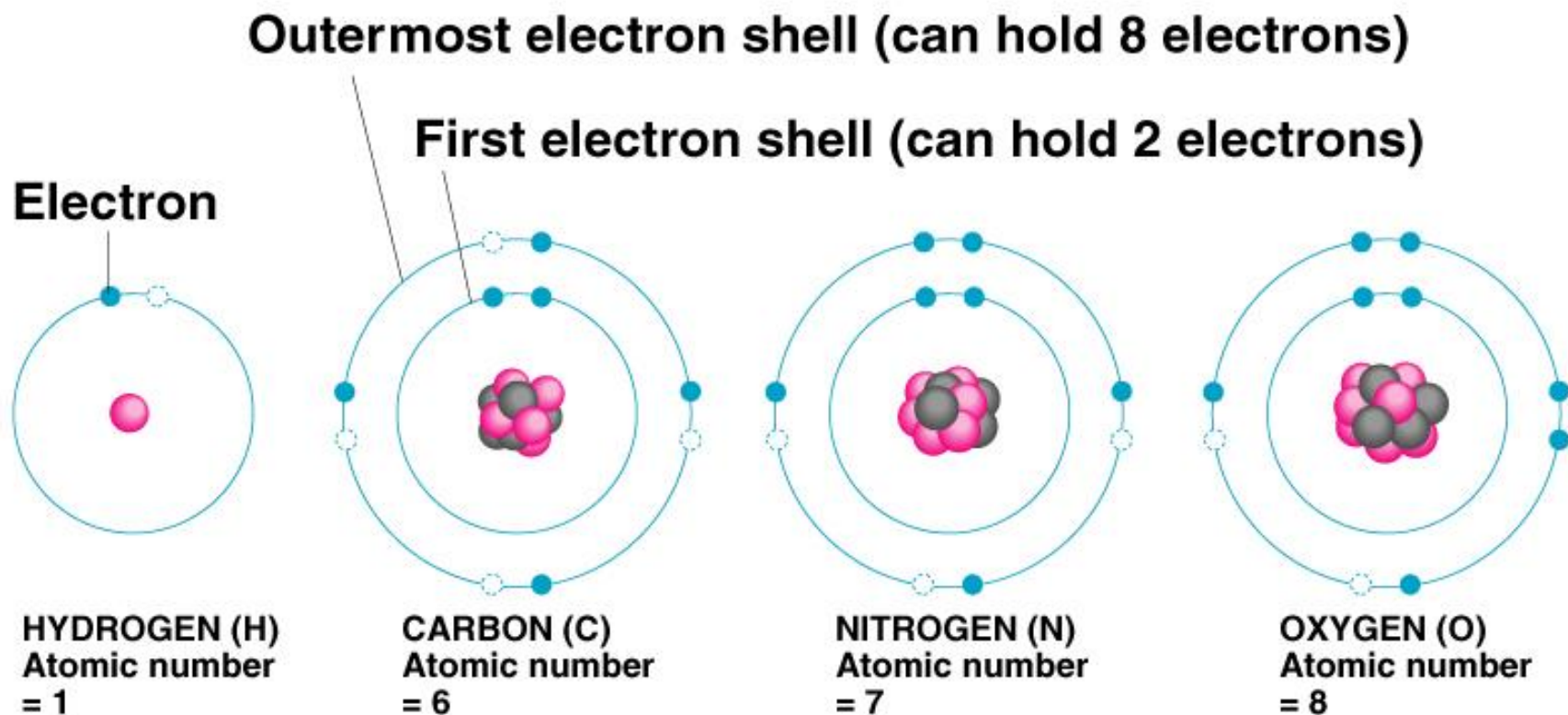
Radiation-induced cell damage

1. Somatic 2. Germline (hereditary)

1. **Carbon dating analysis** : ^{14}C decays into an atom of ^{14}N in 5730 Yrs
2. **Tracer studies (to diagnose image)/Clinical diagnostic/Urinalysis**: Gallium-67 in PET, MRI and I-123 etc.
3. **Radiopharmaceuticals**: Yttrium-90 and Iodine-131 is used as radiopharmaceuticals for the treatment of non-Hodgkin's lymphoma and hyperthyroidism respectively.
4. **Medical Treatments**: Cesium-137 and Cobalt-60 are both used to shrink the size of tumors within the bodies of cancer patients.
5. **Sterilizing**: Sterilization of surgical instruments such as syringes, gloves, clothing and instruments using gamma emitting radionuclides including Cobalt-60, Cs-137 etc.

Isotope	Biological Use	Explanation/Example
^{14}C	Radiocarbon dating	Used to determine the age of fossils and organic remains.
^3H	Tracer in metabolic studies	Incorporated into molecules (e.g., DNA, RNA, lipids) to track synthesis and metabolism.
^{32}P	DNA/RNA labeling	Helps study nucleic acid replication, transcription, and mutation mapping.
^{35}S	Protein studies	Labels sulfur-containing amino acids (methionine, cysteine) to track protein synthesis.
^{13}C	Metabolic pathway analysis	Non-radioactive tracer used in NMR to study metabolic flux and carbon cycling.
^{15}N	Protein and nitrogen cycle studies	Traces protein turnover, nitrogen fixation, and ecosystem nitrogen cycling.
^{18}O	Photosynthesis & water cycle research	Used to trace oxygen in water and CO_2 during photosynthesis and respiration.
$^{99\text{m}}\text{Tc}$ (Technetium-99m)	Medical imaging	Widely used in diagnostic imaging of organs (though medical-biological).

Chemical Bonds



Octet Rule = atoms tend to gain, lose or share electrons so as to have 8 electrons

- ✓C would like to
- ✓N would like to
- ✓O would like to

Gain 4 electrons

Gain 3 electrons

Gain 2 electrons

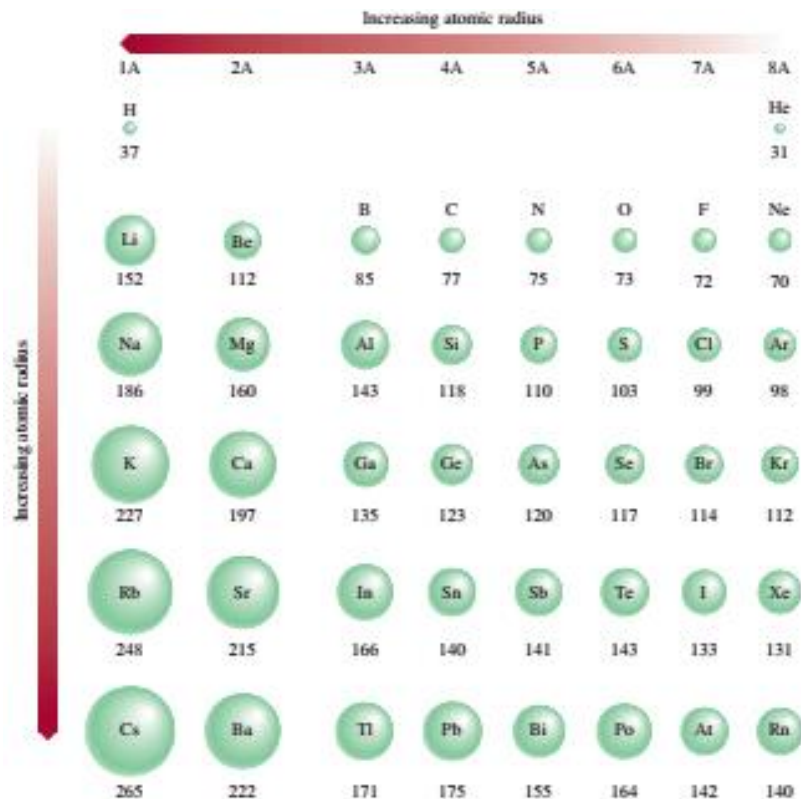
Maximum number of electron
can be accommodated in a
orbit is $2n^2$.

n = number of orbit

K- 2 e; L-8 e; M-18 e; N-32 e

Atomic Radius

A number of physical properties, including density, melting point, and boiling point, are related to the **sizes of atoms**, but atomic size is **difficult to define**. → the **size of an atom** in terms of its **atomic radius**, which is **one-half the distance between the two nuclei in two adjacent metal atoms**.



Ionic Radius

Ionic radius is the radius of a cation or an anion. Ionic radius affects the physical and chemical properties of an ionic compound. For example, the **three-dimensional structure of an ionic compound depends on the relative sizes of its cations and anions.**

- When a neutral atom is converted to an ion, we expect a change in size. If the atom forms an **anion**, its size (or radius) **increases**, because the **nuclear charge remains the same** but the repulsion resulting from the additional electron(s) **enlarges the domain of the electron cloud.**
- Removing one or more electrons from an atom **reduces** electron-electron repulsion but the **nuclear charge remains the same**, so the **electron cloud shrinks**, and the **cation is smaller than the atom**

If we examine **isoelectronic ions**, we find that **cations are smaller than anions**. For example, Na^+ is smaller than F^- . Both ions have the same number of electrons (**10**), but Na ($Z = 11$) **has more protons than** F ($Z = 9$). The larger effective nuclear charge of Na^+ **results in a smaller radius.**

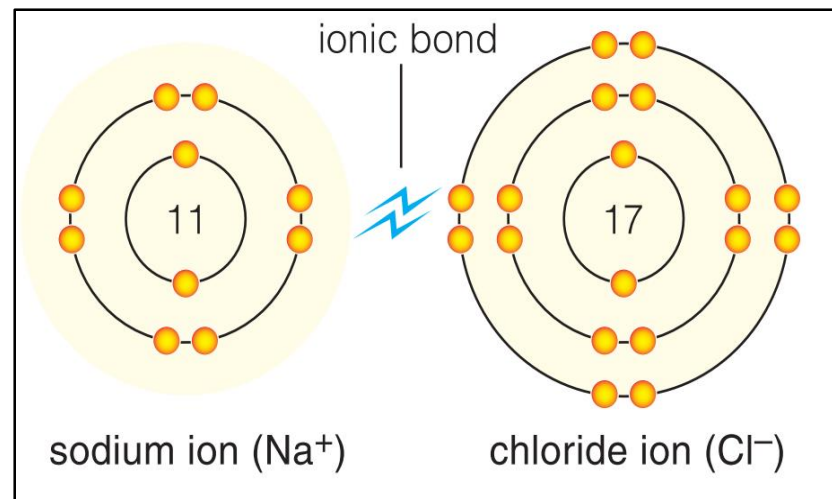
Chemical bonds: an attempt to fill electron shells

A chemical bond is defined as a force that acts between two or more atoms to hold them together as a stable molecule

1. **Ionic bonds**
2. **Covalent bonds**
3. **Hydrogen bonds**
4. **van der Waals interactions:** intermolecular interactions that do not involve covalent bonds or ions.

Ionic Bond

- Transfer of Electron from the outermost shell.
- atom donate e^- and become **cation**.
- another atom receives that e^- and becomes an **anion**.
- An ionic bond is strong.
- Example: NaCl is an ionic compound.



- Ionic compounds result when metals react with nonmetals
- Metals *lose* electrons to match the *number of valence electrons* of their nearest noble gas
- *Positive ions* form *when* the number of electrons are *less* than the number of protons

Formation of Sodium Ion

Sodium atom



e^- in shells **2-8-1**

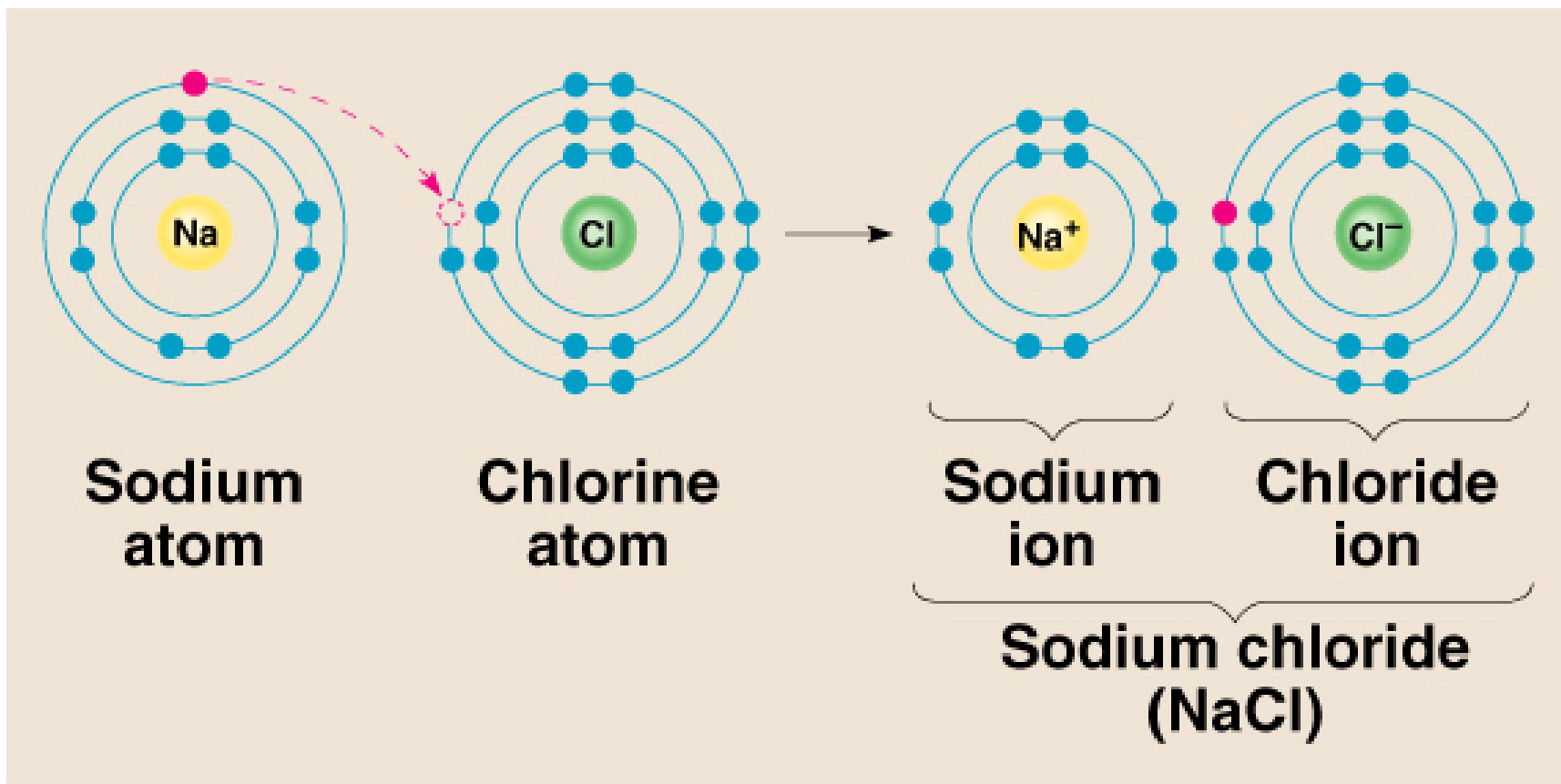
$$\begin{array}{r} 11 p^+ \\ 11 e^- \\ \hline 0 \end{array}$$

Sodium ion



2-8 (= Ne)

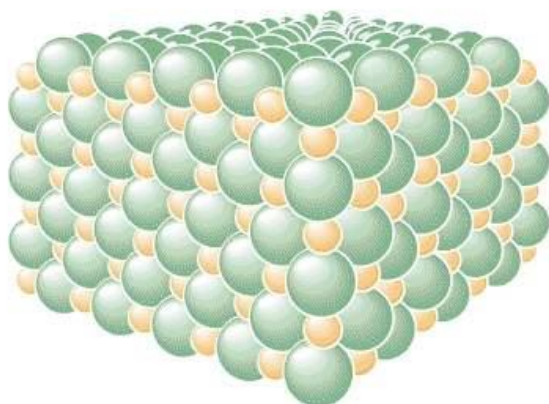
$$\begin{array}{r} 11 p^+ \\ 10 e^- \\ \hline 1^+ \end{array}$$



1). Ionic bond – electron from Na is transferred to Cl, this causes a charge imbalance in each atom. The Na becomes (**Na⁺**) and the Cl becomes (**Cl⁻**), charged particles or ions.



+



Sodium ion (Na^+)



Chloride ion (Cl^-)

Covalent Bond

Chemical bonds are formed by the **sharing of a pair of electrons between atoms**. Between nonmetallic elements of **similar electronegativity**. Examples; O_2 , CO_2 , C_2H_6 , H_2O ,

The **force that holds the atoms together** can still be regarded as arising from the attraction of oppositely charged particles: **nuclei and electrons**.

Bonds in all the polyatomic ions and diatomics are all covalent bonds (H_2 or O_2 or H_2O)

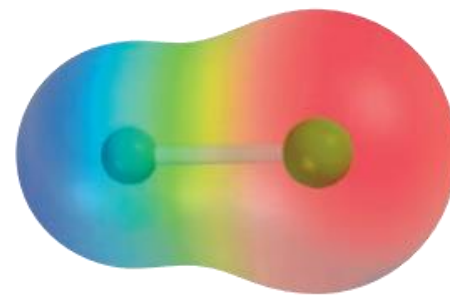
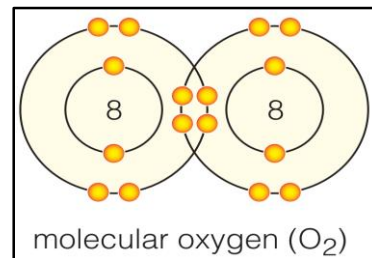
1. Polar Covalent Bonds:

A **covalent bond** in which the **bonding electrons spend more time near one atom** than the other.

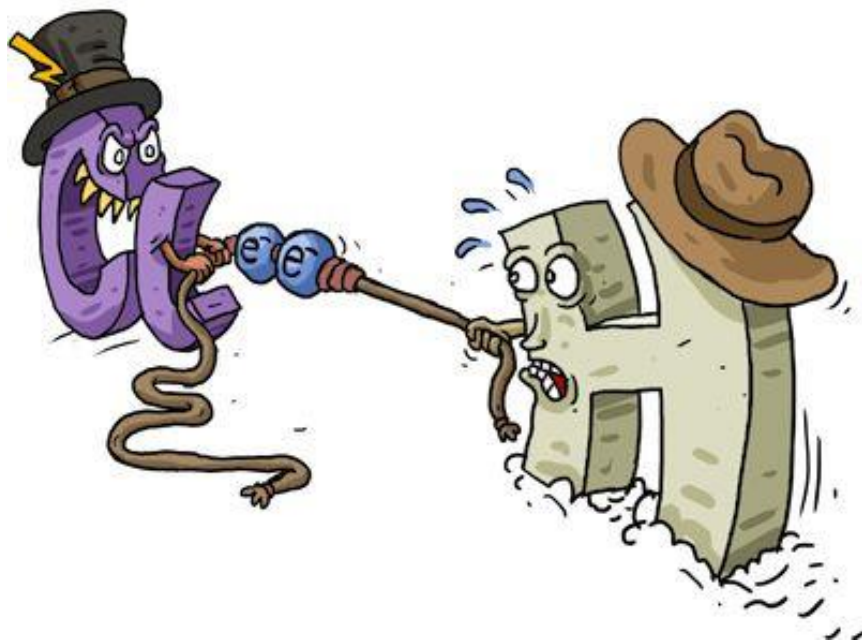
i.e. HCl and HF

In the HF molecule the **electrons spend more time near the F (red) atom**.

The HCl molecule is said to be a **polar molecule**.



Polar Covalent Bond

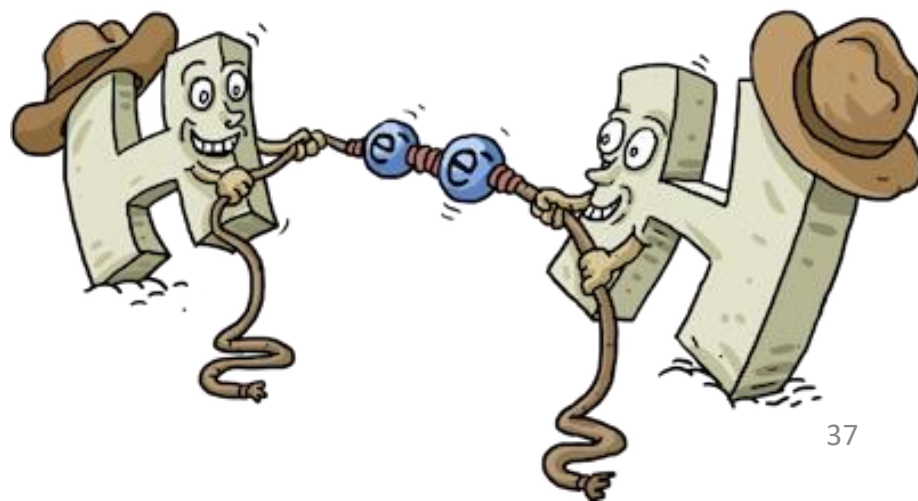


SO.....

Ionic and **covalent** bonds are very different, but how do we predict if a bond will be ionic or covalent?

The **ELECTRONEGATIVITY** of an element helps us understand the difference between ionic and covalent bonding

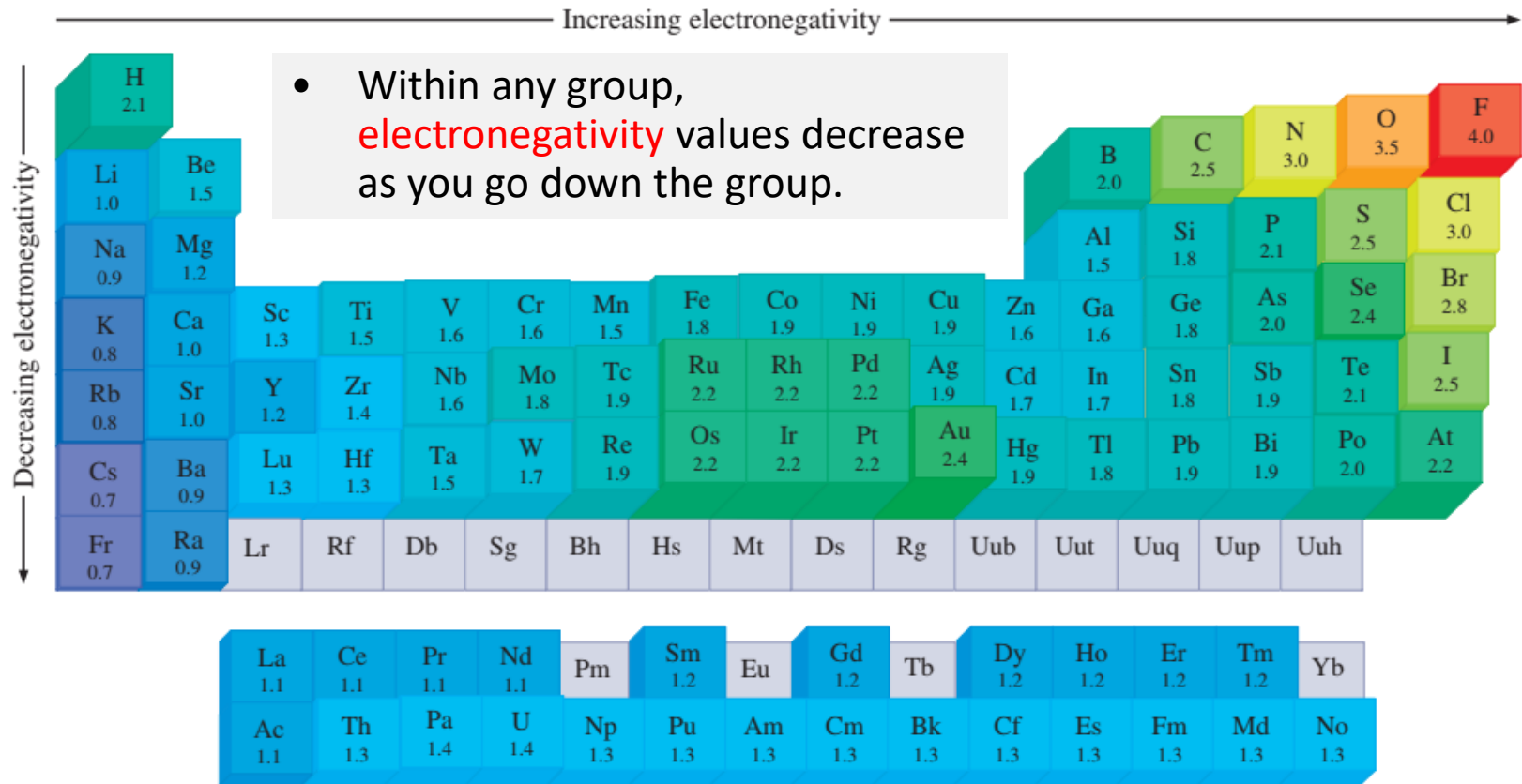
Non-Polar Covalent Bond



Electronegativity

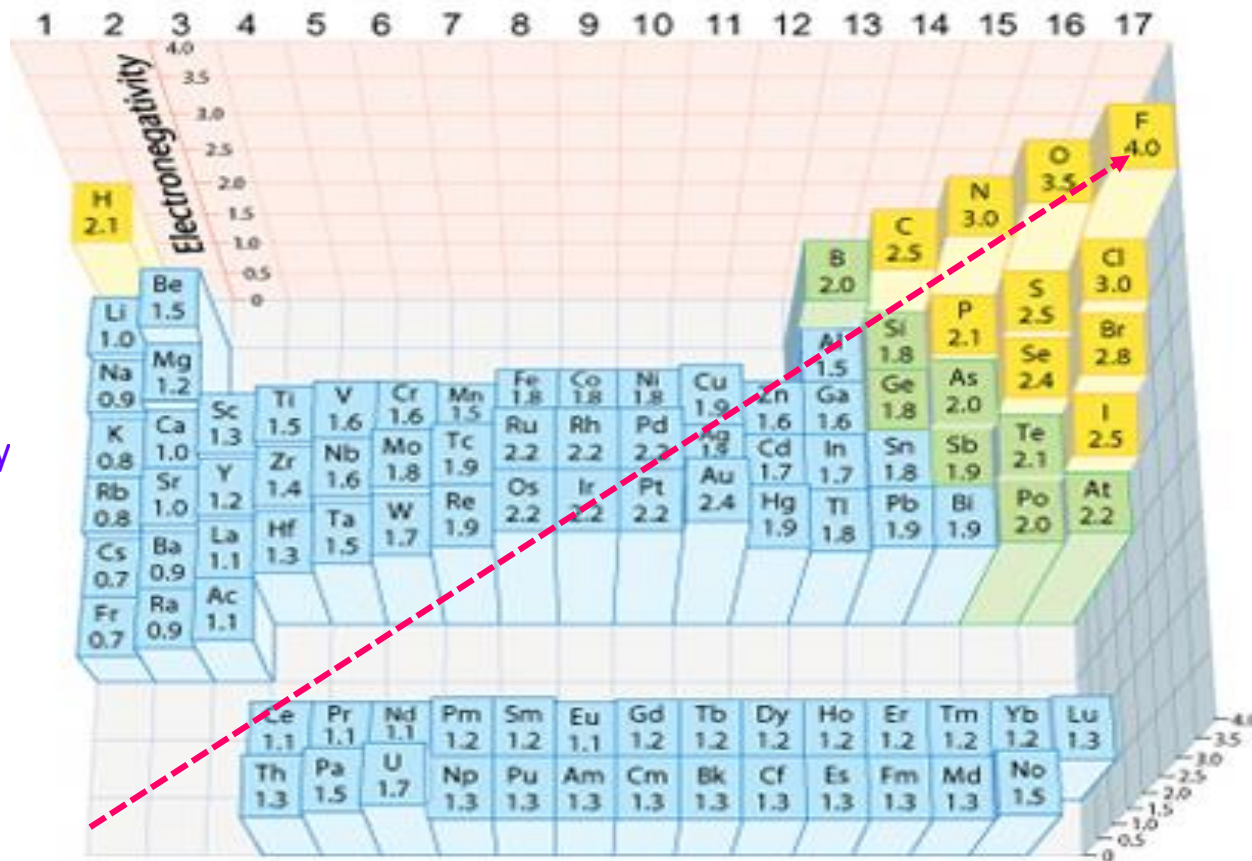
Electronegativity is a measure of the ability of an atom in a molecule **to draw bonding electrons to itself** (the ability of an atom to attract the bonding electrons).

- With only a few exceptions, **electronegativity** values increase as you move from left to right in any period of the periodic table.



That means that the most **electronegative** elements are in the **upper-right** corner of the table.

Every element
has an
electronegativity
value



Francium has the lowest electronegativity 0.7 Fluorine has the highest 4.0

Polar Bonds and Polar Molecules

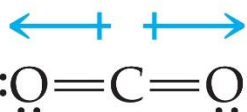
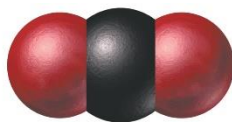
C → 2.5

O → 3.5

Consider carbon dioxide:



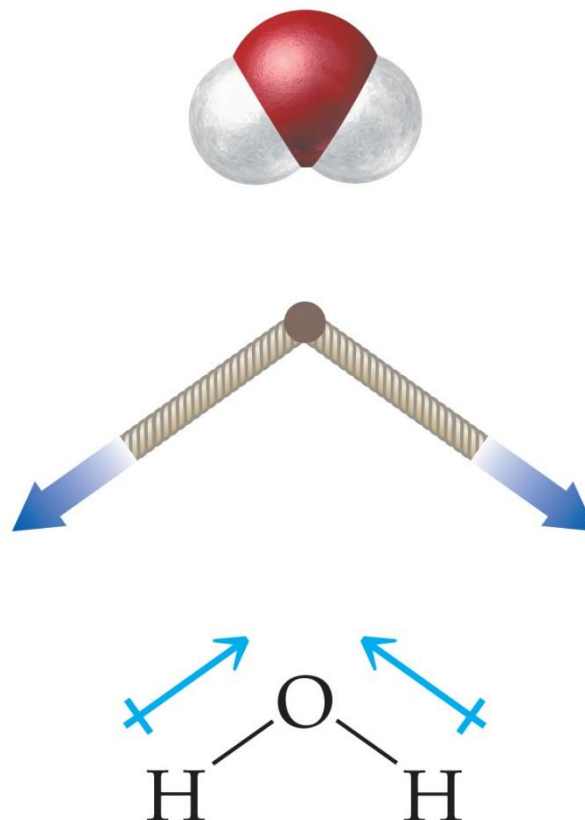
- Each **bond is polar** because the difference in electronegativity between **oxygen and carbon is 1.0**.
- CO₂ has a **linear geometry**, the dipole moment of one bond **completely cancels the dipole moment** of the other, and the **molecule is nonpolar**.



Polar Bonds and Polar Molecules

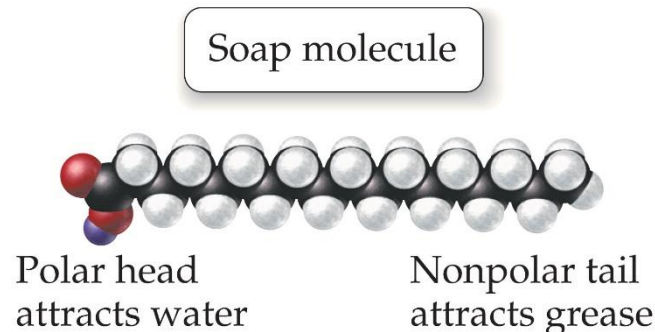
Consider water (H_2O):

- Each *bond is polar* because the difference in electronegativity between *oxygen* and *hydrogen* is 1.4.
- Water has two **dipole moments** that **do not cancel**, and the **molecule is polar**.



Everyday Chemistry- How Soap Works

- After eating a greasy **Biriyani**, your hands are coated with grease and oil. If you try to wash them with only water, they remain greasy. However, if you **add a little soap, the grease washes away. Why?**
- Water molecules are polar, and the molecules that compose grease and oil are nonpolar. As a result, **water and grease repel each other.**
- **One end of a soap molecule is polar**, while the **other end is nonpolar.**



- The **polar head of a soap molecule strongly attracts water molecules**, while the **nonpolar tail strongly attracts grease and oil** molecules.
- **Soap allows water and grease to mix**, removing the grease from your hands and washing it down the drain.

2. NONPOLAR COVALENT BONDS

when electrons are **shared *equally***



Two atoms share one or more pairs of outer-shell electrons.

In a **true covalent bond** electrons are **shared equally**

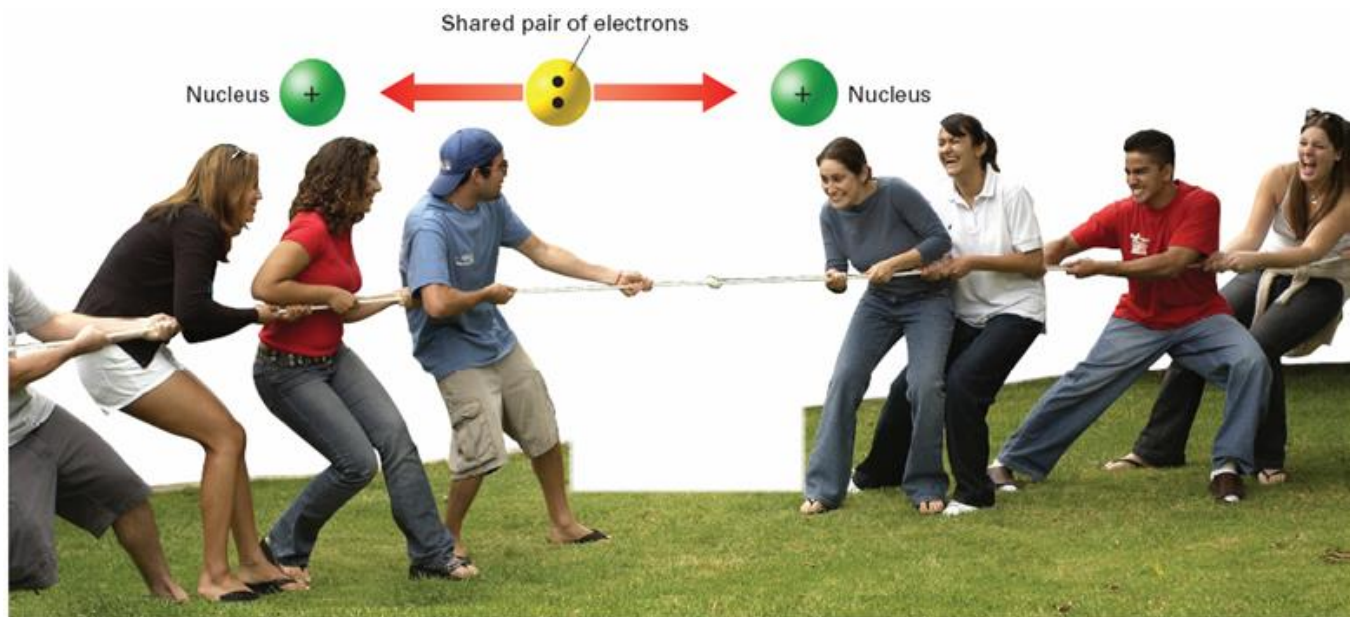


Table: **The Effect of Electronegativity Difference on Bond Type**

Electronegativity Difference (ΔEN)	Bond Type	Example
zero (0–0.4)	pure covalent	Cl_2
intermediate (0.4–2.0)	polar covalent	HF
large (2.0+)	ionic	NaCl

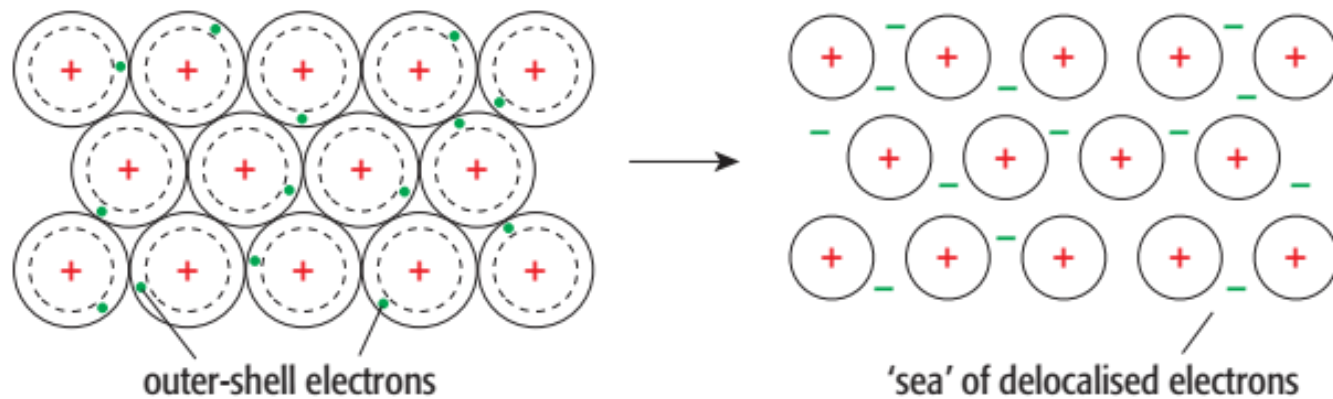
Metallic bond

Metal atoms in a lattice **tend to lose their outer shell electrons** and become positive ions. The outer shell electrons **occupy new energy levels and are free to move throughout the metal lattice**. We call these electrons delocalised electrons (mobile electrons).

The positive ions are held together by the strong electrostatic attraction between their positive charges and the negative charges of the delocalised electrons.

The **strength** of metallic bonding **increases** with:

- **increasing positive charge** on the ions in the metal lattice
- **decreasing size** of metal ions in the lattice
- **increasing number** of mobile electrons per atom.



Most metals have high melting points and high boiling points - A lot of energy is needed to weaken the strong attractive forces between the metal ions and the delocalised electrons. These attractive forces can only be overcome at high temperatures.

Metals conduct electricity- When a voltage is applied to a piece of metal, an electric current flows in it because the delocalised electrons (mobile electrons) are free to move. Ionic solids cannot conduct because neither their electrons nor their ions are free to move.

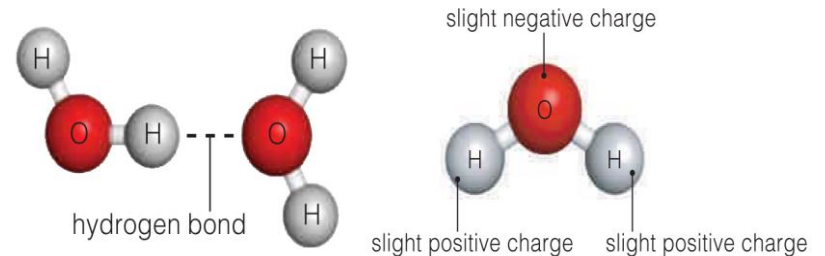
Metals conduct heat- The conduction of heat is partly due to the movement of the delocalised electrons and partly due to the vibrations passed on from one metal ion to the next.

Hydrogen bonds

What do you need?

- one molecule having a **hydrogen atom** covalently bonded to **F, O or N** (the three most electronegative atoms)
- a second molecule having a **F, O or N atom** with an **available lone pair of electrons**.

If only **one** of these conditions is met, you don't get hydrogen bonding.

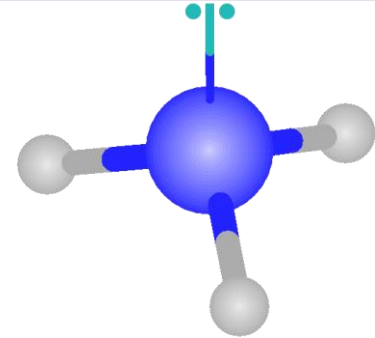


Methane, CH₄ does not have any hydrogen bonds. Carbon is not very **electronegative**, and it has no **lone pairs of electrons** in methane.

Hydrogen bonds

Give me real examples!

Ammonia, NH_3



This does have **one** hydrogen bonds.

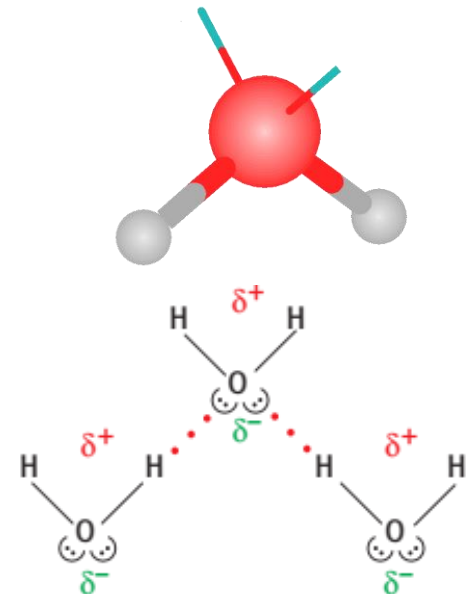
Nitrogen is very **electronegative**, and it has **one lone pair of electrons** in ammonia.



Water, H_2O ...

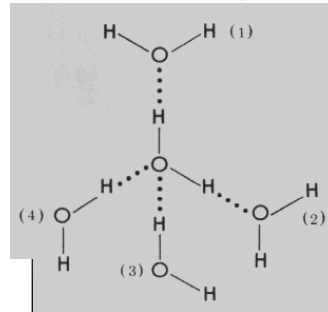
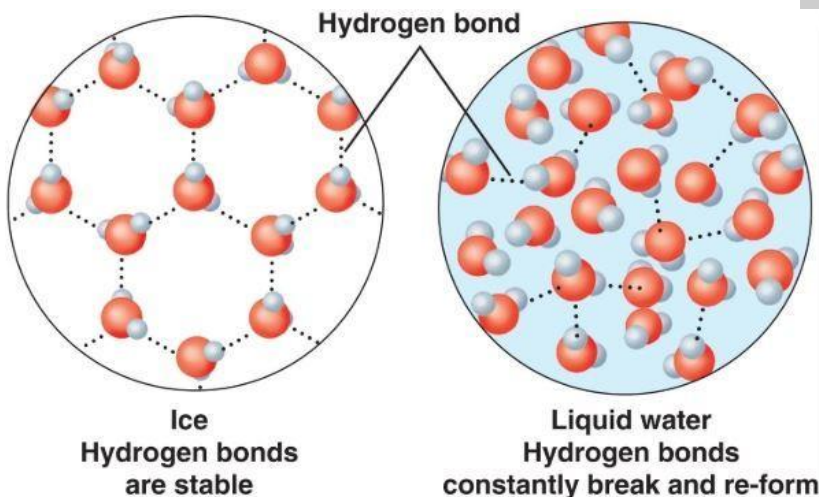
This has not one, but **two** hydrogen bonds.

Oxygen is very **electronegative**, and it has **two lone pairs of electrons** in water.



Ice is less dense than water -

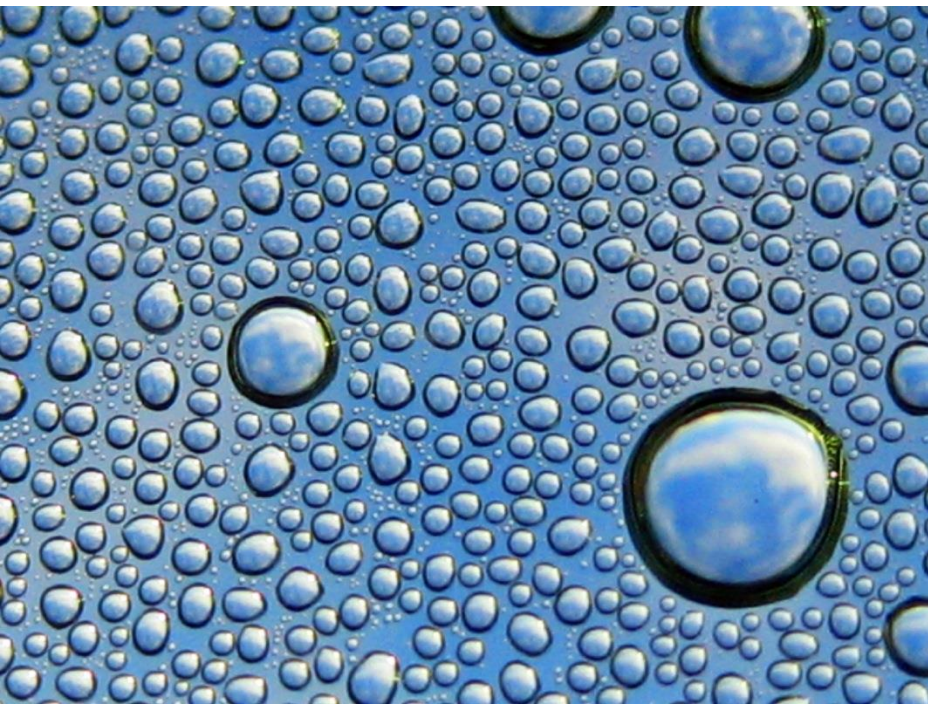
Most solids are denser than their liquids. This is because the molecules are more closely packed in the solid state. In ice, there is a **three-dimensional hydrogen-bonded network of water** molecules. This produces a **rigid lattice in which each oxygen atom is surrounded by a tetrahedron of hydrogen atoms**. This **‘more open’ arrangement**, due to the relatively long hydrogen bonds, allows the **water molecules to be slightly further apart** than in the liquid.



Hydrogen bond – biological system

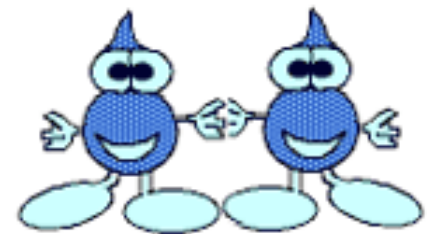
- Human DNA
- Proteins- antibody
- Cellulose
- Nylon
- H₂O absorption by plant
- Tear comes out of eyes
- Water droplet formation- surface tension- capillary action

Water & It's Properties



Water

- Water: **Polar covalent bonds** gives water its **unique properties** that make life possible
- Water is an excellent solvent
 - Hydrophilic substances
 - Hydrophobic substances
- Water is stable in high temperature
 - Why ice floats on water?
- Water is cohesive/adhesive
- Water evaporates



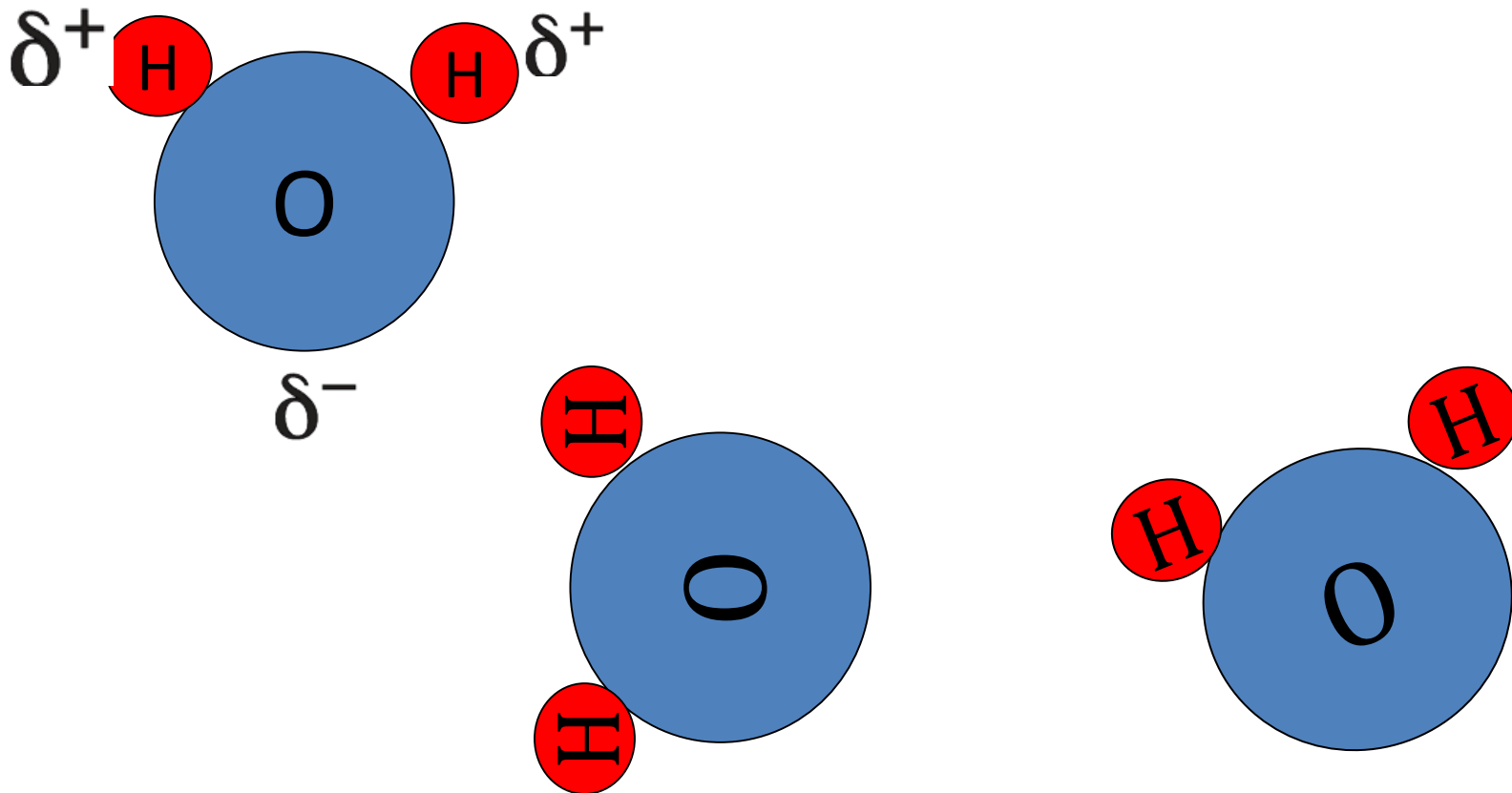
Cohesion



Adhesion

Polar molecule

The **positive** hydrogen ends of 1 molecule are attracted to the **negative** end of the oxygen of another molecule.



Polar molecule

The properties of:

Cohesion, adhesion, specific heat, universal solvent, capillary action, surface tension

ALL HAPPEN BECAUSE WATER IS A POLAR SUBSTANCE!

Surface Tension

- The tightness across the surface of water that is caused by the polar molecules pulling on one another.
- Makes the surface act like a solid



Universal Solvent

- Water is called the "universal solvent" because it is capable of dissolving more substances than any other liquid. This is important to every living thing on earth.

- Remember:

SOLVENT = the thing doing the dissolving

SOLUTE = the thing that dissolves away

- Other polar molecules can be dissolved by water.



Why is water so good at dissolving things?

Because water is a polar molecule and is shaped like a wedge, it is able to break up substances into smaller pieces (dissolve).

Non-polar Substances

- Non-polar substances DO NOT dissolve in water.
- Nonpolar molecules are molecules that do not have any electrical charges or partial charges. Nonpolar molecules are made of atoms that have a similar value of electronegativity.
- Can you think of some examples of non-polar substances?

Hydrocarbon,
fats, waxes
etc

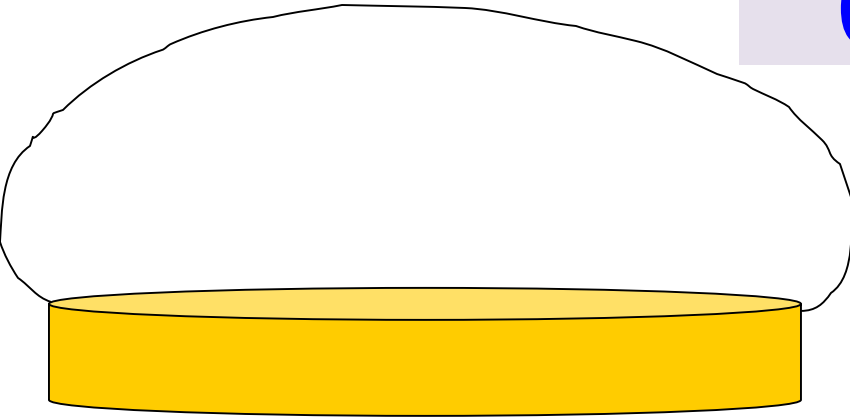


*oil and water don't
mix*

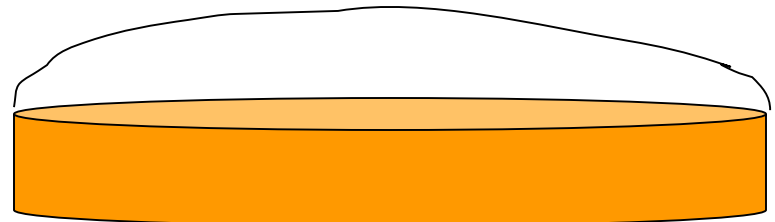
COHESION

Cohesion – is the attractive force between water molecules.

Water molecules
like to stick to
each other.



Rubbing alcohol
doesn't like to
stick to itself as
much as water.



Adhesion

- Because water is a polar molecule, it is attracted to other substances.
- **Adhesion** – occurs when molecules of water are attracted to other substances.



Cohesion

Water molecules
like to stick to
each other.

Adhesion

Water molecules
like to stick to
other things.

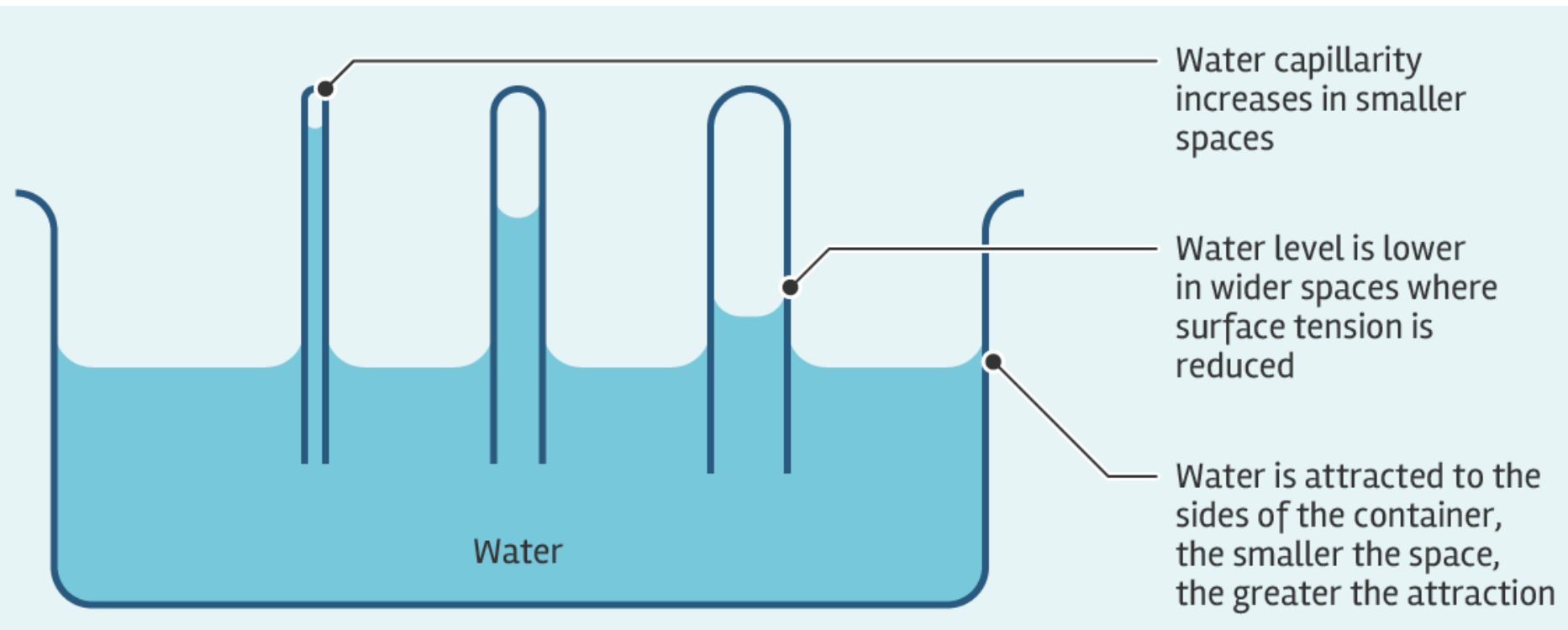


BOTH ADHESION AND COHESION ARE DEMONSTRATED

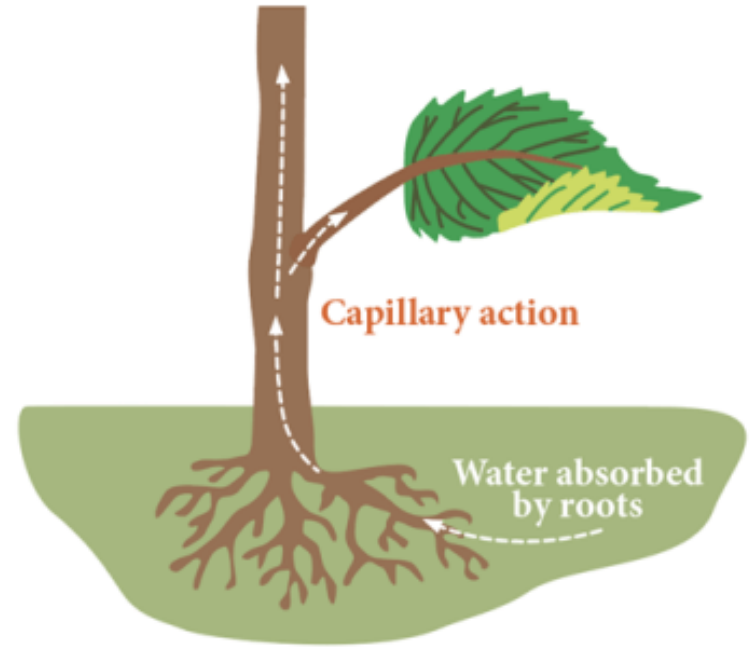
What is Capillary Action?

Capillary action is when water travels in a narrow space due to the forces of cohesion and adhesion against the force of gravity.

- The combined force of attraction among water molecules and with the molecules of the surrounding material.
- Cohesion + Adhesion



Uses of capillarity effect on day to day basis:

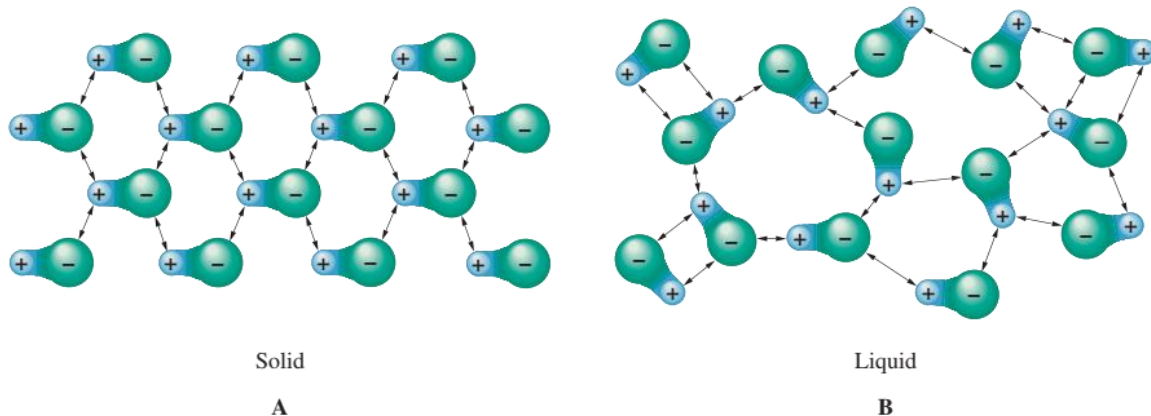


Van der Waals's forces

The term van der Waals forces is a general term for those intermolecular forces that include **dipole–dipole and London forces**.

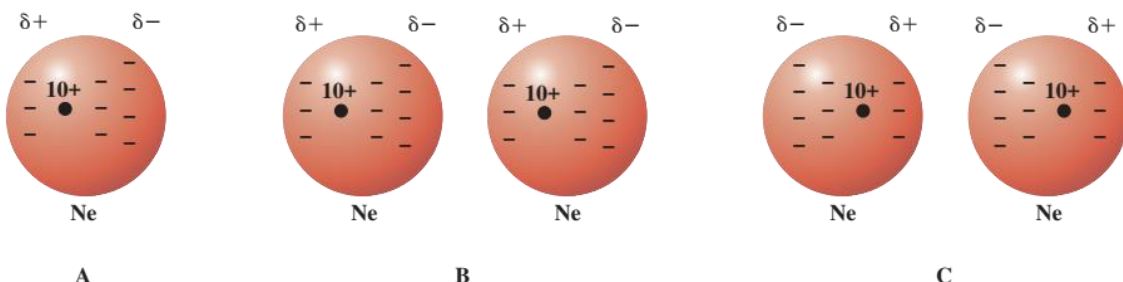
Dipole–Dipole Forces- is an attractive intermolecular force **resulting from the tendency of polar molecules to align themselves** such that the positive end of one molecule is near the negative end of another.

The alignment of **polar molecules in the case of HCl**. This alignment creates a net attraction between molecules. This attractive force is partly responsible for the fact that hydrogen chloride, a gas at room temperature, **becomes a liquid when cooled to -85°C** . At this temperature, HCl molecules have slowed enough (that is, have a low enough kinetic energy) for the **intermolecular forces to hold the molecules in the liquid state**.



London (Dispersion) Forces- In nonpolar molecules, there can be no dipole–dipole force. Even in hydrogen chloride, calculations show that the dipole–dipole force accounts for less than 20% of the total energy of attraction.

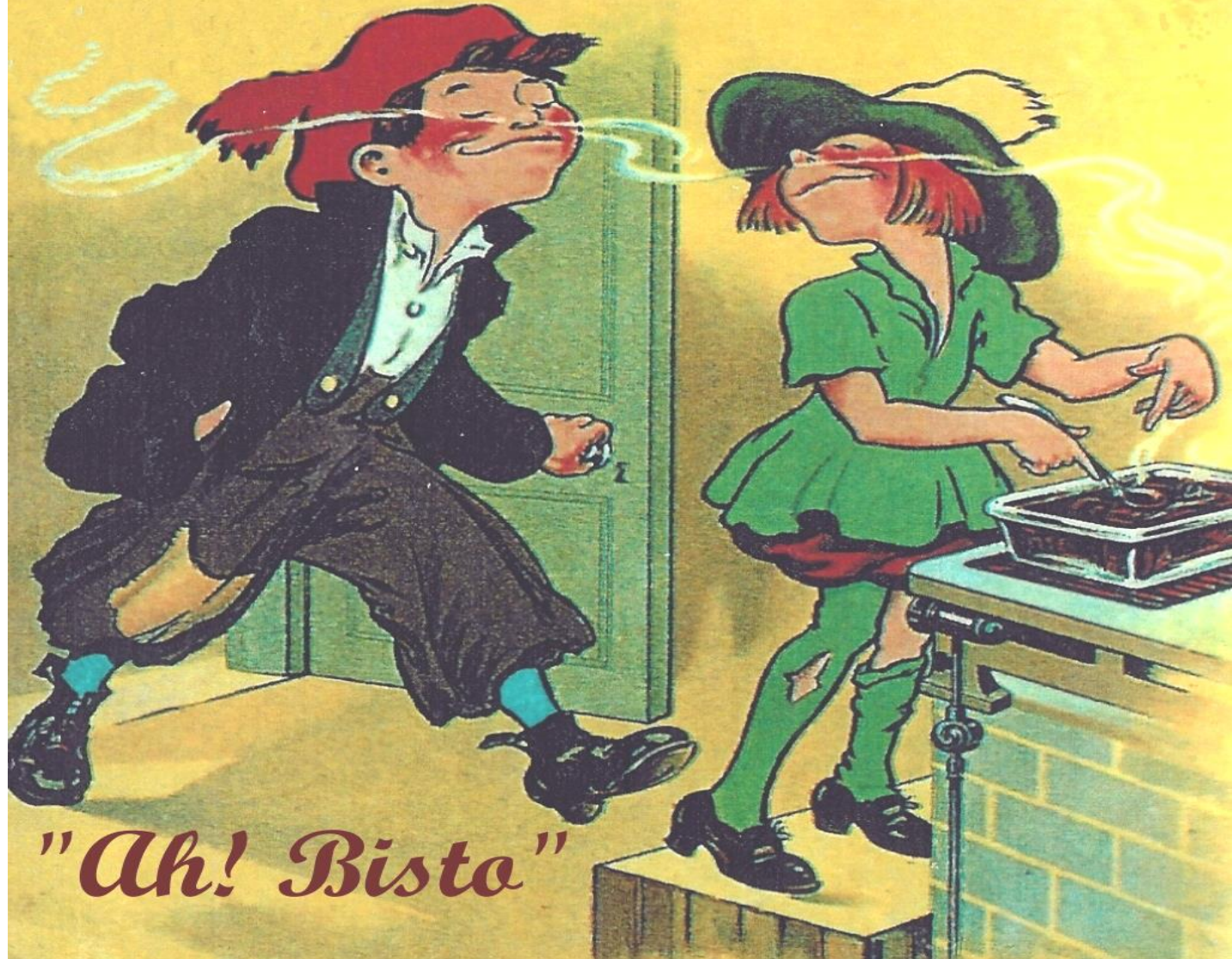
What is the nature of the additional attractive force?



F and Cl are **gases** at room temperature, Br is a **liquid**, and I is a **solid**. (HI is **gaseous**)

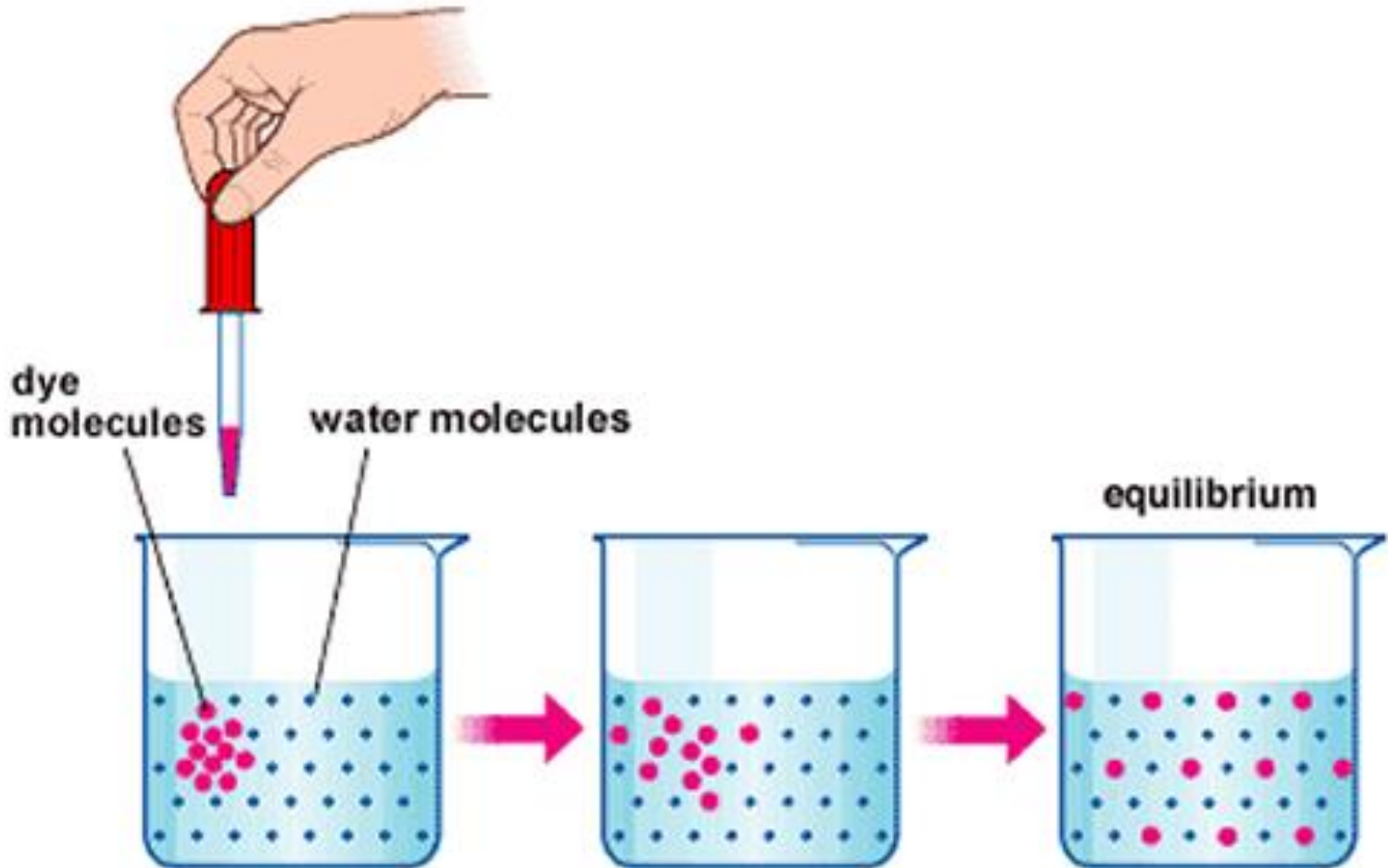
The instantaneous dipoles of the atoms tend to change together, maintaining a net attractive force. **London forces tend to increase with molecular mass.** This is because molecules with larger molecular mass usually have more electrons, and London forces increase in strength with the number of electrons (increased polarizability of molecules with larger, more dispersed electron clouds.)

Diffusion



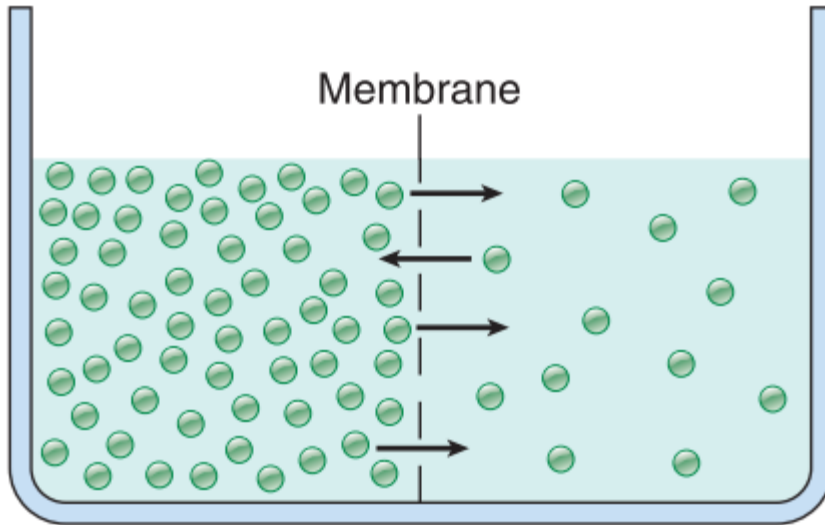
Diffusion

The process by which molecules spread from areas of high concentration, to areas of low concentration.

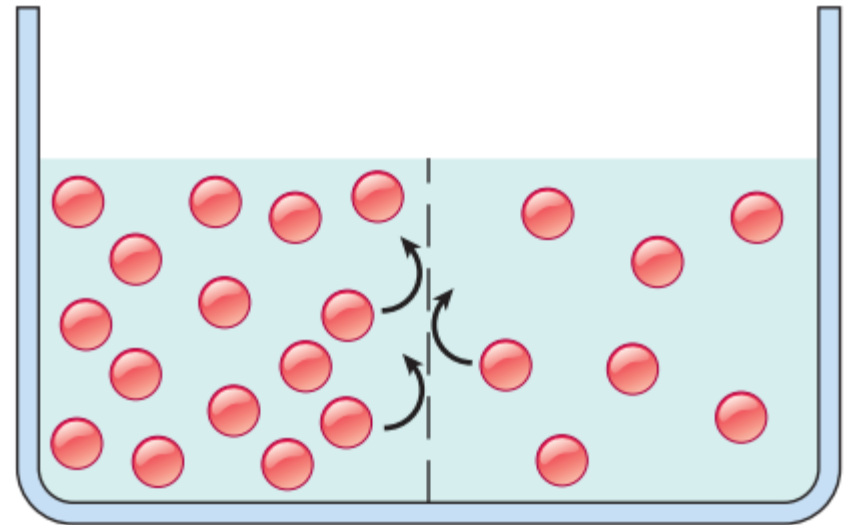


Diffusion

If a substance can permeate the membrane:



If the membrane is impermeable to a substance:



(a) Diffusion occurs

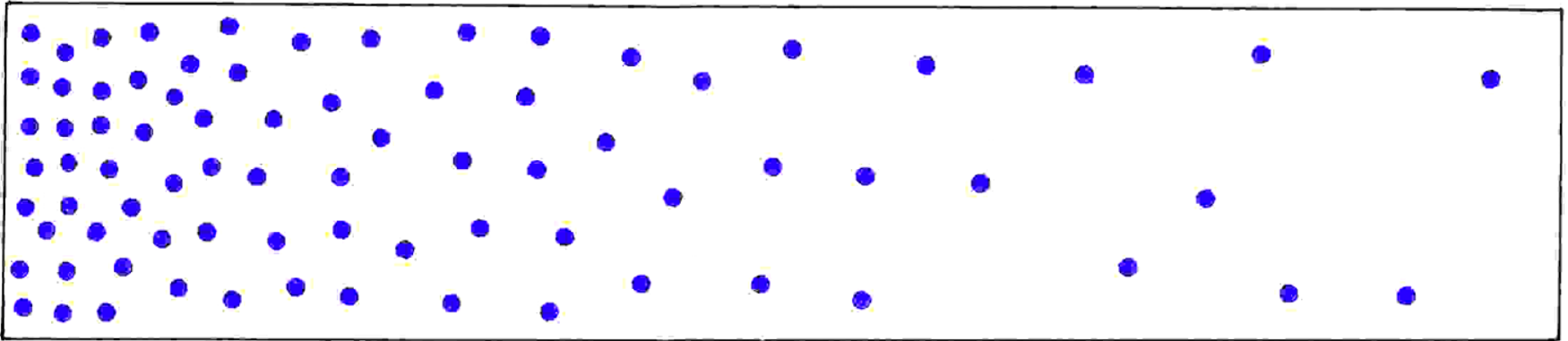
(b) No diffusion occurs

KEY

- = Penetrating solute
- = Nonpenetrating solute

Diffusion gradient

The molecules are more densely packed on the left and so they tend to diffuse into the space on the right. This is a **diffusion gradient**

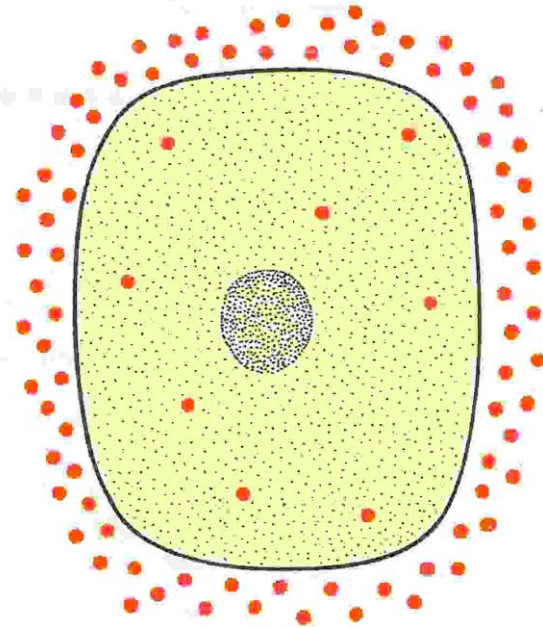


A diffusion gradient

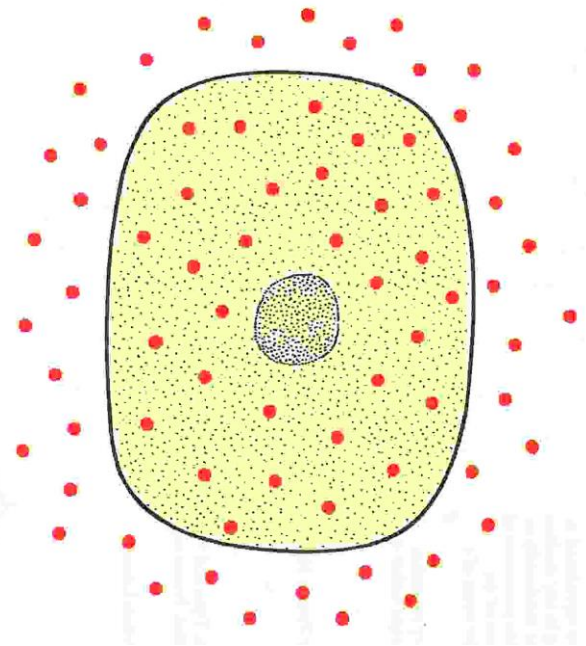


Diffusion of oxygen into a cell

The concentration of oxygen molecules is greater outside the cell than inside

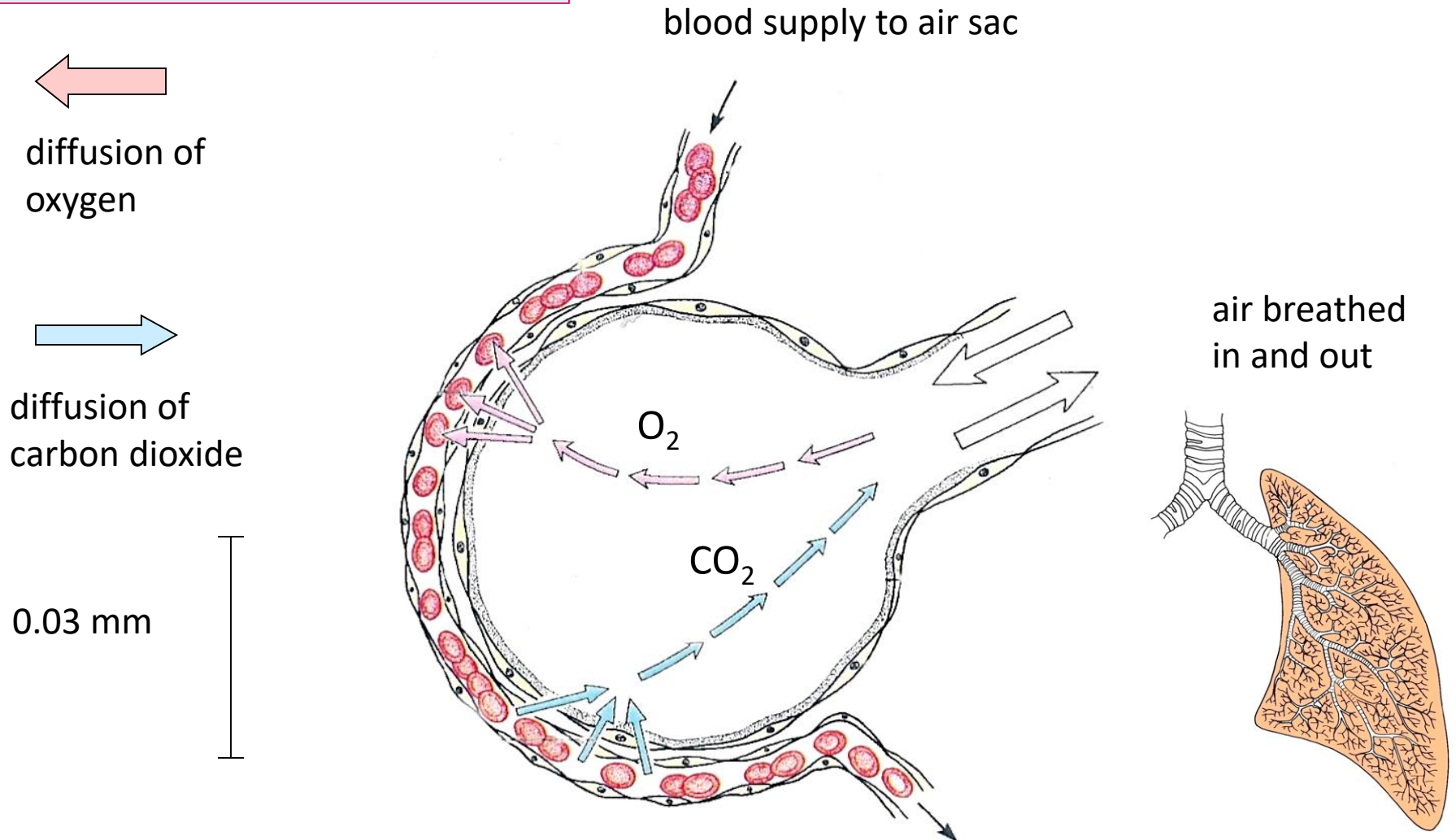


So the oxygen molecules diffuse into the cell



Because the cell is using up oxygen, the concentration of oxygen inside the cell is always lower than the concentration outside. The diffusion gradient is maintained → So oxygen continues to diffuse in

A single air sac

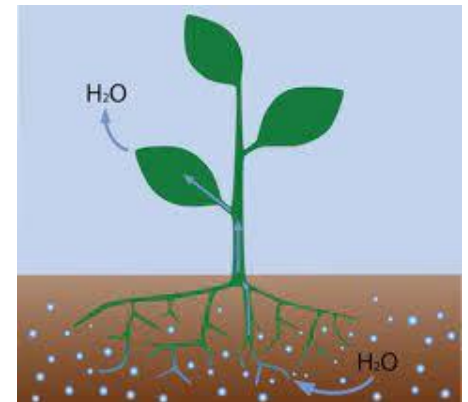
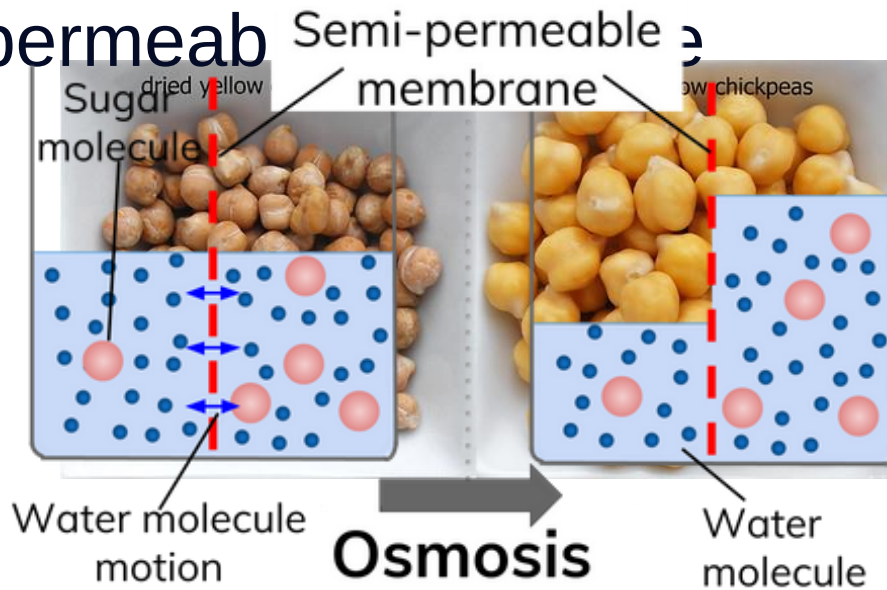


In mammals, birds, reptiles and amphibia, oxygen and carbon dioxide are exchanged by diffusion in the lungs.

In fish, this exchange of gases takes place by diffusion through the gills

Osmosis

Osmosis is a process by which the molecules of a solvent pass from a solution of low concentration to a solution of high concentration through a semi-permeable membrane.

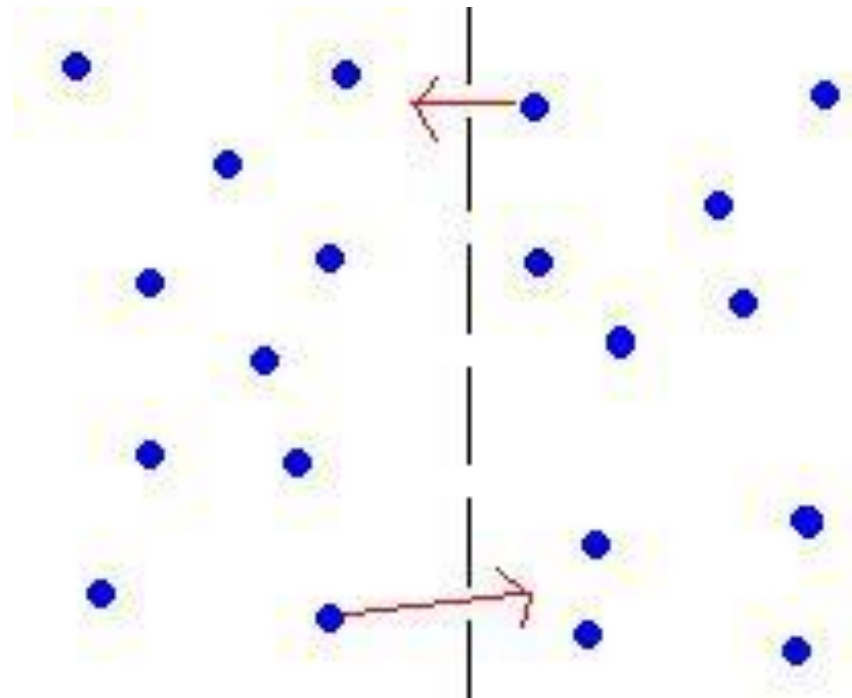


water enters the root cells by osmosis

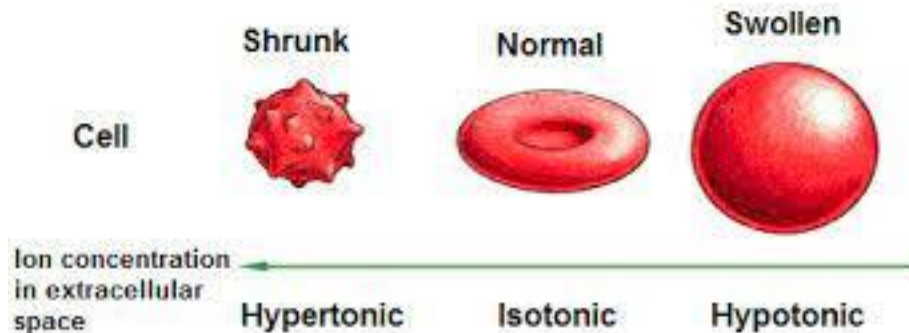
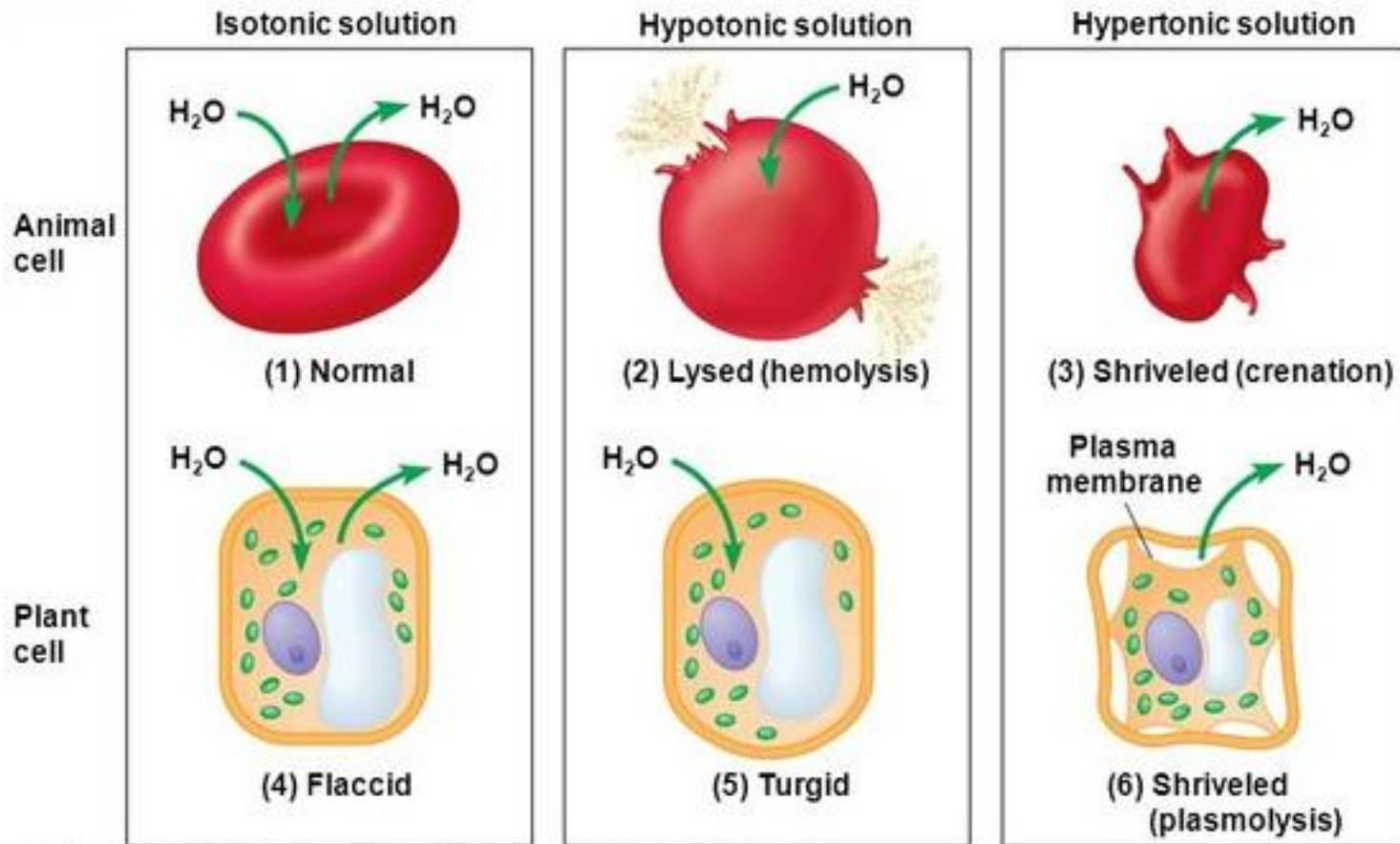


Osmotic effect leads to the secretion of water into the intestinal lumen resulting in a softening of the stool.

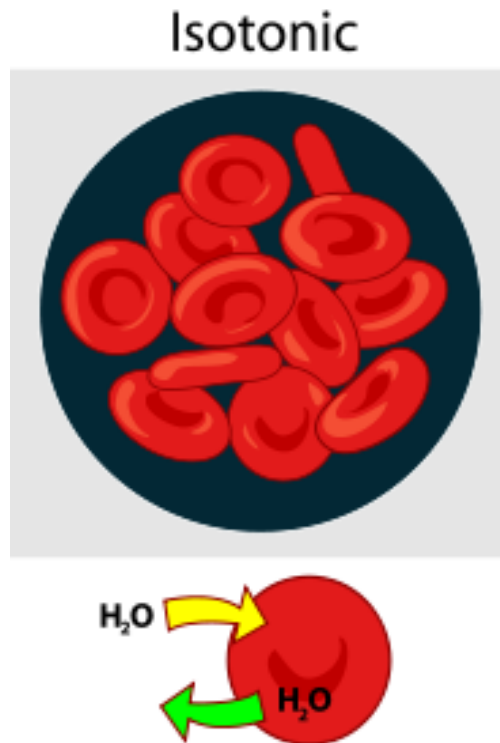
Distilled water separated by a partially-permeable membrane:



Water molecules are moving from one side of the membrane to the other but there is NO NET OSMOSIS

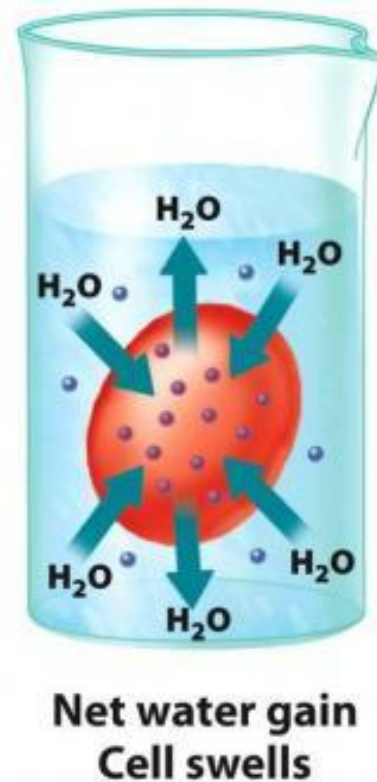
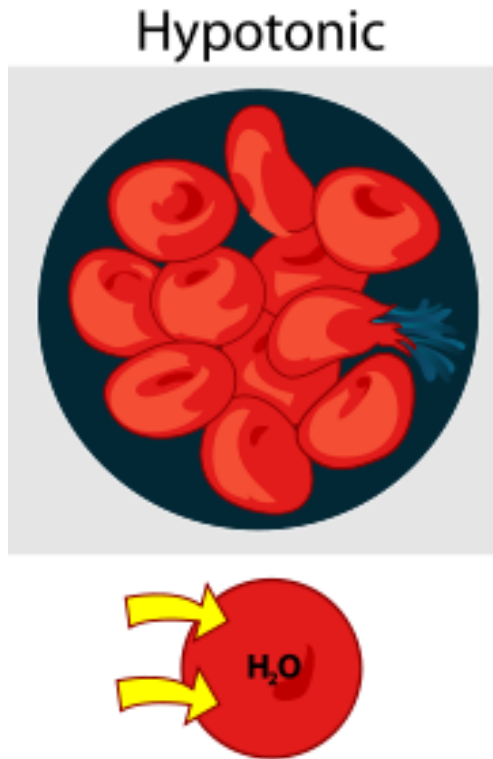


Physiological saline is 0.9% NaCl.



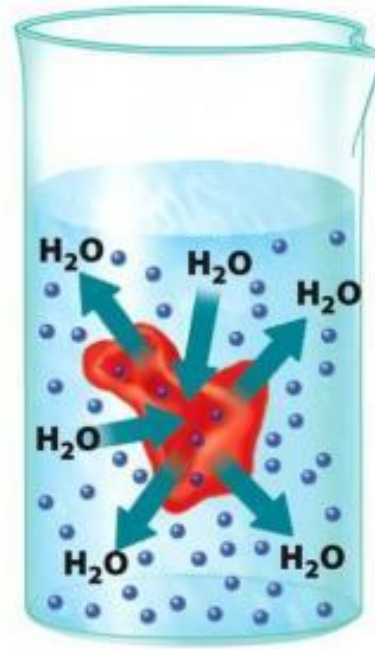
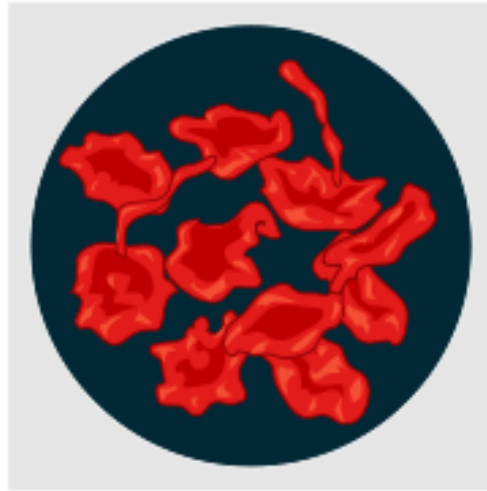
No net loss or gain

An isotonic solution is one in which outside solute concentration is same as the solute concentration of a cell. In this case the cell neither swells nor shrinks because there is no concentration gradient for water across the cell membrane



A hypotonic solution has a lower concentration of solutes outside the cell than inside the cell. In an attempt to balance the concentrations of solutes inside and outside the cell, water will rush into the cell, and can cause it to burst

Hypertonic



Net water loss
Cell shrinks

A hypertonic solution is one with a higher concentration of solutes outside the cell than inside the cell. When a cell is immersed into a hypertonic solution, the tendency is for water to flow out of the cell in order to balance the concentration of the solutes.

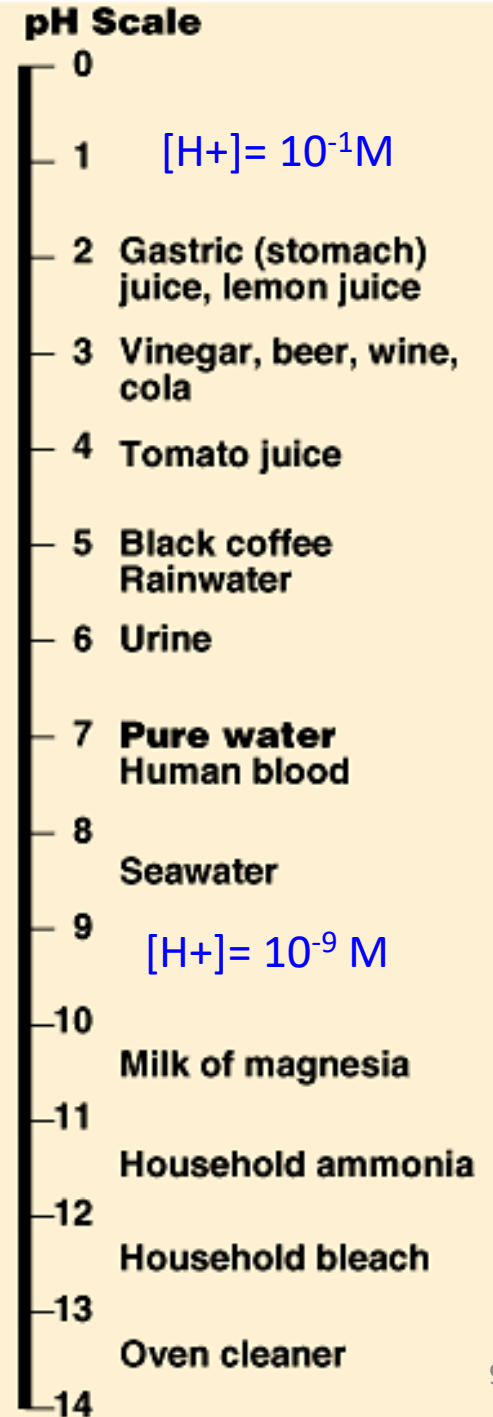
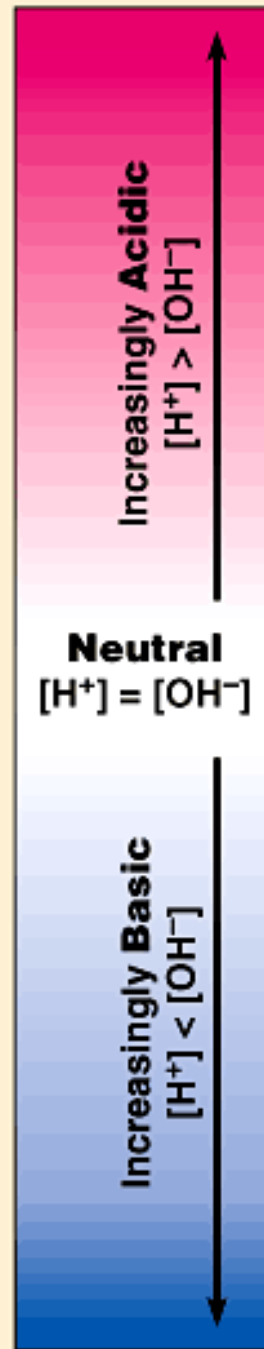


Theory	Acid	Base
Arrhenius	A hydrogen-containing substance that produces hydrogen ions in aqueous solution	A hydroxide-containing substance that produces hydroxide ions in aqueous solution
Brønsted–Lowry	A proton (H⁺) donor	A proton (H⁺) acceptor
Lewis	Any species that will bond to an unshared pair of electrons (electron pair acceptor)	Any species that has an unshared pair of electrons (electron pair donor)

pH

$$\text{pH} = -\log [\text{H}^+]$$

*Log scale means 10X
change per unit!*



BUFFER SOLUTION

A **buffer solution** is one which maintains its pH fairly constant even upon the addition of small amounts of acid or base. [a buffer solution resists (or buffers) a change in its pH]

Two common types of buffer solutions are :

1. a **weak acid together with a salt of the same acid** with a strong base. These are called acid Buffers e.g.,
 $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$.
2. **a weak base and its salt** with a strong acid. These are called Basic buffers. e.g., $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$.

Body's Acid-Base Balance

**First line of
defense against
pH shift**

**Chemical
buffer system**



**Bicarbonate
buffer system**

**Phosphate
buffer system**

**Protein
buffer system**

**Second line of
defense against
pH shift**

**Physiological
buffers**

**Respiratory
mechanism
(CO₂ excretion)**

**Renal
mechanism
(H⁺ excretion)**